

Refinancing Inequality and Monetary Transmission Channel

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Abstract

This paper documents substantial mortgage refinancing heterogeneity across income groups and quantifies its implications for the transmission of monetary policy. Using a loan-level panel data, I show that refinancing inequality appears in both the *intensity* and *timing* of refinancing: the bottom income quintile refinances at two-thirds the rate of the top quintile and faces markedly longer delays, even when refinancing is financially beneficial. These gaps persist even after controlling for income- and time-varying measures of lender credit tightness. The associated unrealized savings are large, exceeding 8.4% of monthly income for the bottom quintile. I develop a structural mortgage refinancing model that decomposes refinancing frictions into two distinct channels—*inattention* and *hassle cost*—and allows both frictions to vary by income group. The model matches the observed refinancing inequality, unlike homogeneous-friction benchmarks. Counterfactual experiments show that reducing inattention frictions for low-income households would amplify the effects of a policy rate cut by 7.1% in consumption over five years. Refinancing inequality dampens monetary transmission, and mitigating inattention strengthens the real effects of rate cuts.

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1 Introduction

Mortgages play a key role in household finance due to their scale and prevalence. One of the most prominent features of the U.S. mortgage market is that it is dominated by long-term fixed-rate mortgages. As of February 2024, about 92% of all outstanding mortgages are fixed-rate. This market feature emphasizes the role of refinancing as a transmission channel of monetary policy. For example, when the Federal Reserve lowers interest rates, mortgagors can benefit from refinancing into lower-rates, thereby accessing more liquidity to increase their consumption. At the same time, studies have identified significant friction in refinancing (Keys et al. 2016; Andersen et al. 2020; Gerardi et al. 2023; Zhang 2022) and it could significantly impair the effectiveness of monetary policy (Beraja et al. 2019; Defusco et al. 2019; Defusco and Mondragon 2020).

However, this paper documents substantial refinancing heterogeneity across income groups, even within the presence of average refinancing frictions. Low-income households engage significantly less in refinancing than high earners, a phenomenon referred to as “refinancing inequality.” Figure 1 plots mortgage rates and prepayment activity over time. During periods of declining interest rates, significantly higher prepayment activity is observed, suggesting prepayment activity is highly driven by refinancing to take advantage of lower borrowing costs. A notable feature of the figure is that the prepayment response varies substantially across income quintiles, even though all borrowers face the same aggregate interest-rate environment. A notable feature of the figure is that the prepayment response varies substantially across income quintiles. Higher-income borrowers responding far more strongly than lower-income borrowers.¹ This casts an important implication on monetary policy that the refinancing channel may not effectively serve households likely to be under liquidity constraints, described as “wealthy hand-to-mouth” by Kaplan and Violante (2014).² Despite this importance, refinancing inequality across income groups remains understudied in the literature. Furthermore, connecting the refinancing frictions to consumption response and quantifying their impact on monetary policy have not been explored in previous studies.

¹Income quintiles follow the same definition as in the main analysis: borrowers are assigned based on nationwide annual mortgage income distributions at the time of loan origination, constructed from the Freddie Mac data.

²Specifically, the Heterogeneous Agent New Keynesian model (HANK) literature indicates that the lower the income relative to consumption (high average propensity to consume), the higher the marginal propensity to consume (MPC) (eg. Auclert 2019). For more direct evidence, the 2022 American Community Survey (ACS) reports that over 98.9% of mortgagors with an annual income of \$2,000 or less are ‘cost-burdened,’ spending more than 30% of their income on housing costs. In contrast, 70.2% of households with an annual income between \$35,000 and \$50,000, 45.5% of households with an income between \$50,000 and \$75,000, and only 11.2% of households with an income above \$75,000 are cost-burdened. This clearly

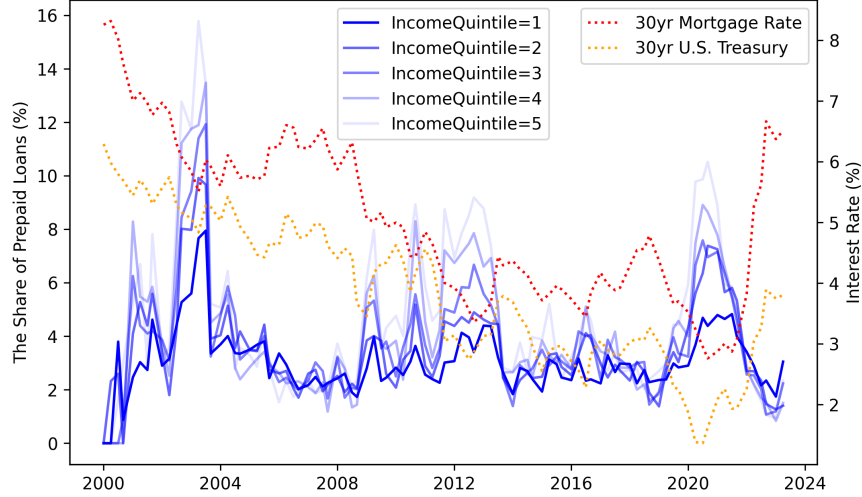


Figure 1: Prepayment Activity by Income Quintile

This paper proceeds in three steps. First, using loan-level panel data, I document substantial refinancing inequality across income groups, not only in the likelihood of refinancing but also in the timing of refinancing once households become in the money. Second, I show that this inequality extends to the transmission of monetary policy, with low-income households responding significantly less to interest rate cuts. Finally, I develop a structural mortgage refinancing model with heterogeneous attention and execution frictions to interpret these patterns and quantify their macroeconomic implications.

I begin by documenting refinancing inequality using loan-level panel data from 2000 to 2022. The analysis reveals substantial refinancing inequality in both the *intensity* and *timing* of refinancing. When households are in the money, those in the lowest income quintile exhibit monthly refinancing probabilities that are roughly two-thirds of those in the top quintile. Decomposing refinancing behavior by the gap between the contracted mortgage rate and the market rate, I find that high-income households consistently display significantly higher refinancing probabilities than low-income households. Importantly, the inequality is most pronounced in the in-the-money region, indicating that even when refinancing is advantageous, low-income borrowers are far less likely to act. Beyond differences in refinancing probability, I provide novel evidence that refinancing inequality also appears in its *delays*. Measuring the share of loans that refinance after becoming in the money, I show that higher-income house-

demonstrates that as absolute income increases, the proportion of income spent on housing decreases. Therefore, it can be reasonably inferred that lower-income households are likely to have less available liquidity.

holds refinance much more promptly and therefore capture financial gains earlier than their lower-income counterparts.

A natural concern is that these income-quintile gaps might be driven by lender-side underwriting. I therefore control for year-by-income mortgage rejection rates across all U.S. markets and a quarterly survey-based index of lending standards, and the refinancing gaps remain large and statistically significant. Moreover, robustness checks that stratify borrowers into credit-score quantiles show that, within any given credit-score group, refinancing inequality across income groups persists.

To quantify how much lower-income households miss out on potential gains from refinancing, I estimate the economic magnitude of forgone savings, which are particularly large for low-income borrowers. By matching refinanced (old) and refinancing (new) loans, I estimate the reduction in monthly mortgage payments due to refinancing. Households in the bottom and second-lowest quintiles could have saved approximately 8% of their monthly income had they refinanced when it was optimal to do so. This amount is substantial not only as a share of income but also because it translates into a persistent increase in monthly disposable income.

I also present previously undocumented evidence of heterogeneity in the refinancing response to monetary policy shocks. Using high-frequency identification of monetary policy surprises, I find that low-income households respond much less to interest rate cuts than high-income households. Following a one-percentage-point reduction in the policy rate, households in the top income quintile refinance roughly 18% of their outstanding mortgages within seven quarters, whereas those in the bottom quintile refinance only about half as much. These results demonstrate that refinancing inequality persists not only in levels but also in the transmission dynamics of monetary policy.

Beyond the empirical evidence, I develop a structural mortgage refinancing model to interpret the observed refinancing inequality and to quantify its macroeconomic implications. The key innovation of the model is to decompose refinancing frictions into two conceptually distinct channels—an inattention channel and an execution (hassle-cost) channel—and to allow both frictions to vary systematically across income states. This structure makes it possible to separately quantify the contribution of each channel to refinancing behavior. Attention heterogeneity is disciplined using a parsimonious parametric specification consistent with empirical patterns, while execution frictions are estimated nonparametrically across income groups. By allowing refinancing frictions to differ by income and by channel, the model successfully reproduces the observed refinancing inequality. In contrast, benchmark models in the existing literature

that impose homogeneous refinancing frictions substantially overpredict refinancing among low-income households and underpredict it among high-income households.

The model further demonstrates that these frictions have meaningful implications for monetary policy transmission. A counterfactual experiment—raising the attention capacity for low-income households to the level observed among high-income borrowers—shows that a one-percentage-point interest rate cut would generate an additional 7.0%-8.2% increase in aggregate consumption over five years, depending on the past interest rate paths. This amplification arises because higher attention enables more borrowers to refinance promptly when rates fall, thereby strengthening the pass-through of monetary policy to household cash flows and consumption. Since attention-related frictions are potentially mitigable through targeted policy interventions, the results highlight a realistic avenue through which monetary easing can be made more effective.

Related Literature This paper contributes to two key areas of literature. The first explores refinancing frictions, where numerous studies since the financial crisis have shown that households often fail to refinance optimally ([Agarwal et al. 2016](#); [Keys et al. 2016](#); [Andersen et al. 2020](#); [Gerardi et al. 2023](#)). Common explanations for these sub-optimal decisions include closing costs, financial illiteracy, inattention, and racial discrimination. My analysis primarily uses the same data and methods as [Agarwal et al. \(2023b\)](#), who were the first to identify refinancing inequality and analyze income-level differences in refinancing.

Building on this foundation, this paper sheds light on several previously unexamined dimensions of refinancing inequality. First, I link refinancing inequality directly to the transmission of monetary policy—an aspect not explored in their study—by examining heterogeneous IRFs across income groups and by developing a structural refinancing model that quantifies the resulting amplification effects. Second, while [Agarwal et al.](#) focus on differences in refinancing intensity, I show that refinancing inequality also manifests in its timing: lower-income households refinance with significantly longer delays even after becoming in-the-money. Third, I substantially expand the empirical scope beyond the COVID period and generalize the pattern of refinancing inequality. Using a longer time series and a more robust measure of refinancing incentives, I show that the inequality persists across the full distribution of rate incentives and remains present even after controlling for home equity, credit tightness, and other borrower characteristics.

The second strand of literature relates to the refinancing channel of monetary policy. A large body of research has highlighted that this channel significantly impacts the

transmission of monetary policy (Campbell and Cocco 2003; Bhutta and Keys 2016; Di Maggio et al. 2017; Agarwal et al. 2018; Chen et al. 2020; Abel and Fuster 2021; Agarwal et al. 2023a). Furthermore, a number of previous studies have identified notable heterogeneity in this channel. The heterogeneity has been attributed to factors such as housing equity (Beraja et al., 2019), the path of past interest rates (Berger et al. 2021; Eichenbaum et al. 2022), the type of mortgage contract (Calza et al. 2013; Di Maggio et al. 2017), borrower's age (Wong, 2021), inaction (Byrne et al., 2023), and regulation changes (Defusco et al. 2019; Defusco and Mondragon 2020).

However, this literature has not examined how income-based heterogeneity shapes the refinancing channel of monetary policy. This omission is notable, because income is not only one of the most salient dimensions of economic inequality but is also directly linked to households' marginal propensity to consume. As a result, income-group variation in refinancing behavior is likely to matter disproportionately for monetary transmission. Motivated by this gap, this paper provides the first systematic evidence that refinancing inequality across income groups has a quantitatively meaningful effect on the pass-through of interest rate cuts to household consumption.

This paper briefly reviews the data and methodology in [section 2](#), then examines the average pattern of refinancing inequality over the entire period in [section 3](#) and provides a back-of-the-envelope analysis of potential savings that could have been gained by refinancing. In [Section 4](#), the focus shifts to analyzing the refinancing dynamics in response to interest rate changes. Finally, [section 5](#) develops a mortgage refinancing model with frictions to rationalize the empirical findings. The model successfully replicates the observed refinancing inequality and quantifies how relaxing these frictions amplifies the consumption response to interest rate cuts.

2 Institutional Background and Data

2.1 Effect of Income on Refinancing

The potential mechanisms by which income can influence refinancing can be outlined in three main ways. First, there is the friction caused by low liquidity and the presence of closing costs for low-income households. Closing costs typically include fees for the loan application, appraisal, title insurance, and other administrative services. It usually amounts to 2-5% of the loan balance, which can be a significant barrier to refinancing for households with low liquidity. If the borrower does not have immediate liquidity, it is usually possible to roll the closing costs into a new loan, though this presents a trade-off with the interest rate. The tradeoff entails deciding whether to pay

higher upfront closing costs for a lower interest rate, reducing monthly payment over the loan term, or to opt for lower closing costs with a higher interest rate, resulting in higher monthly payments and total interest paid. Appendix [Figure A.2](#) shows an example of the trade-off between upfront cost and interest rate. In this example, it can be observed that for the same property and loan, both cash-out (equity extraction) and non-cash-out refinancing options are associated with a lower interest rate when higher upfront costs are incurred.

Secondly, there is the Payment to Income (PTI) ratio regulation. In the United States, the PTI level is applied as a soft borrowing constraint but is incorporated into the loan approval process by setting a standard. For example, the standard PTI level for a conventional loan is 36%, and a higher PTI level can only be approved if exceptional conditions such as a low LTV and a high credit score are met. As can be seen in actual data in [Table A.6](#), low-income households exhibit higher PTI at loan origination compared to high-income households. Therefore, low-income households are more likely to hit the PTI limit than high-income households given the same distribution of idiosyncratic income shocks.

Lastly, there is the association with inattention. As shown by Andersen et al., inattention, which is a significant cause of observable suboptimal refinancing, is positively correlated with income. Consequently, through behavioral issues, different income groups may exhibit varying refinancing tendencies.

2.2 Data Overview

The primary dataset used in this study is the Freddie Mac loan-level panel data, which covers loans originating from 2000 to 2022. This dataset provides comprehensive information on both loan issuance and performance.³ The issuance data includes a rich set of loan and borrower characteristics at the time of origination. Key variables include the initial loan balance, loan term, interest rate, credit score, Loan-to-Value (LTV) ratio, Payment-to-Income (PTI) ratio, ZIP code, and estimated income. These variables allow for detailed analysis while controlling for factors influencing refinancing decisions, and I restricted sample to fixed-rate mortgages. In addition to issuance data, the performance data offers monthly observations of loans, which include the remaining balance, current interest rate, amortization, and prepayment status. This panel data

³The Freddie Mac data set represents approximately 20%-30% of the total U.S. mortgage market, with the notable exclusion of “jumbo” loans, which are larger than the conforming loan limits set by the Federal Housing Finance Agency every year.

enables the tracking of loan performance over time, facilitating a dynamic analysis of refinancing behavior.

The income for each loan can be back-calculated using Freddie Mac’s issuance data, using the original interest rate, loan amount, maturity, and PTI information. Based on the distribution of this derived income of each origination year, the loans were then divided into quintiles.

To complement the Freddie Mac data, this study also utilizes data from the Home Mortgage Disclosure Act (HMDA) from 2000 to 2022.⁴ The HMDA data consists of detailed records on individual mortgage applications and originations collected by financial institutions. The data include a wide array of information, such as the loan amount, type of loan, property location, borrower characteristics, and the outcome of the application. I matched Freddie Mac to HMDA using common variables to find corresponding loans, and used information from HMDA for analysis.

Detailed summary statistics of the final panel used in the analysis can be found in Appendix [subsection A.1](#).

2.3 Identifying Refinancing Loans

The main challenge when attempting to identify refinancing activity from the Freddie Mac loan panel is it only reports prepayment activity without a reason. As prepayment of mortgages could happen due to various other reasons than refinancing, such as moving or trading of houses, identifying refinancing is a key step for this analysis. I match a prepaid loan to a new loan also funded by Freddie Mac originated within a 45-day window of the closure of the prepaid loan, following [Agarwal et al. \(2023b\)](#).⁵

However, during the matching process, the restricted regional data of the publicly available Freddie Mac panel posed a significant challenge. The Freddie Mac data only provides three-digit ZIP codes for properties, which cover an excessively large area, leading to numerous duplicate matches and making unique matching infeasible. To address this issue, I matched the Freddie Mac data with HMDA data, utilizing the finer regional index at the census tract level provided by HMDA. Given the potential

⁴The Home Mortgage Disclosure Act (HMDA) requires financial institutions to maintain and annually disclose data about home purchases, home purchase pre-approvals, home improvement, and refinance applications. The HMDA data captures the majority of mortgages in the U.S., accounting for about 92 percent of originations nationally in 2017, according to the Consumer Financial Protection Bureau.

⁵ The 45-day window is based on the institutional background of the rate lock period. A rate lock is an agreement between the borrower and the lender that guarantees a specified interest rate for a predetermined period while the mortgage application is being processed. The 45-day period is a common duration within this range. Thus, when a loan is prepaid for refinancing, a new loan is highly likely to be originated within this window.

for bias in the matching processes, I performed a series of data validation checks, as displayed in the Appendix [subsection A.3](#), and found no significant errors in the matched sample compared to the population.

2.4 Identifying In-the-money Loans

The decision for households to refinance is determined by the two types of incentives: first, the cash-out motive due to liquidity needs in response to idiosyncratic shocks, and second, the demand for reducing financing costs through lower mortgage rates. The first refinancing incentive is known to be insensitive to interest rates, while the second incentive is directly affected by monetary policy.⁶ Since the focus of this study is to analyze the differences in the transmission of the benefits of accommodative monetary policy across income groups, the second incentive, the motive for interest rate reduction, should be emphasized. Therefore, distinguishing whether there is a financial benefit from lowering mortgage rates during refinancing is one of the key identifications. An in-the-money option refers to an option that provides a financial benefit when exercised. Therefore, a fixed-rate mortgage becomes an in-the-money loan when refinancing is financially advantageous.

$$\text{In-the-money if } c_{it} - m_t > \theta_{it} \quad (1)$$

The financial benefit of refinancing arises when the borrower’s current mortgage rate exceeds the rate available on a newly originated loan, that is, when the loan is “in the money.” Formally, a loan is identified as in the money when the contract rate $c_{i,t}$ is higher than the prevailing market rate $m_{i,t}$ by more than the threshold $\theta_{i,t}$, as shown in [Equation 1](#).

In the baseline specification, I adopt the simple rule of setting $\theta_{i,t} = 0$, assuming that borrowers refinance whenever the contracted mortgage rate exceeds the market rate. Under this benchmark, a loan is considered in the money if its coupon is above the rate available on newly originated loans with similar characteristics (ZIP code, credit score, LTV, income, and origination quarter). The prevailing market rate $m_{i,t}$ is estimated each quarter as the median coupon among newly refinanced loans within these borrower–loan characteristic bins, following [Defusco and Mondragon \(2020\)](#).

As a robustness check, I also employ a more conservative identification rule based on the Agarwal-Driscoll-Laibson gap (hereafter *ADL gap*), where the threshold

⁶ [Chen et al. \(2020\)](#) focused on the cash-out motive of refinancing and suggested that refinancing decisions of low-liquidity households are insensitive to interest rate changes.

$\theta_{i,t} = \theta_{i,t}^{ADL}$ corresponds to the optimal refinancing threshold derived from [Agarwal et al. \(2013\)](#). This approach incorporates refinancing costs, expected exogenous prepayment, maturity, and interest rate volatility, thereby tightening the condition under which a loan qualifies as in the money. Although more restrictive, this ADL-based rule yields similar qualitative patterns of refinancing inequality. The detailed formula of the ADL gap is explained in [Appendix C](#).

3 Evidence of Refinancing Inequality

3.1 Average Refinancing Probability

I begin by identifying varying refinancing probability across income groups using a linear probability model (LPM).⁷ In this regression estimation, I analyze only the loan observations that were in-the-money. The reason for this is that the focus of this study is on whether interest rate reductions lead to heterogeneous benefits across income groups, so refinancing unrelated to interest rate reductions is not of interest.⁸ Specifically, I used the following specification:

$$Refi_Ind_{it} = \alpha + \sum_{q=2}^5 \beta_q \cdot IncomeQuintile_{iq} + \Gamma' \cdot X_{it} + v_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \quad (2)$$

Refi_Indicator is a binary variable that equals 1 if the loan is refinanced and 0 otherwise. j denotes income quintiles 2 through 5. X_{it} represents a set of control variables, and $v_{z(i)}$ and $\lambda_{v(i)}$ are ZIP code and vintage year-quarter fixed effects, respectively. Therefore, the β_j means the refinancing propensity of income quintile j compared to the bottom quintile, conditional on being in-the-money.

One concern is that the difference in refinancing probabilities across quintiles could be driven by other factors. To control for loan and borrower characteristics that are influential in refinancing, I used a rich set of control variables of credit score, loan age, loan balance to income ratio, ZIP code, and original vintage year, following literature. Also, the control variables were included as categorical variables to allow for non-parametric regression.⁹

⁷The LPM assumes a linear relationship between the dependent variable and the explanatory variables, but since I allowed for nonlinearity by using category variables for income quintiles, I found no significant differences compared to the Logit results.

⁸For example, households who face idiosyncratic income shocks would refinance to extract equity regardless of interest rate, such as [Chen et al. \(2020\)](#) documented.

⁹I used categorical variables for LTV, FICO, loan age, loan balance, and estimated LTV.

However, existing research has not controlled for two other important variables that affect refinancing by income level. First, as highlighted in the literature by [Beraja et al. \(2019\)](#), the equity distribution channel could drive heterogeneous refinancing activities by income group. For example, a low-income borrower would be more likely to purchase a home in a low-income area. If low-income areas are more susceptible to a decline in home prices during an economic downturn, the likelihood that the low-income borrower will become underwater increases, resulting in a lower ability to refinance. Such dynamic changes cannot be controlled by regional fixed effects. To address this, I controlled for the equity channel by estimating the loan-to-value (LTV) ratio for all loans on a monthly basis. Starting with the LTV and loan size at origination, I derived the collateral value and tracked home values monthly using the Zillow ZIP code-level house price index. By combining the estimated home value with the observed remaining loan balance from the loan performance data, I was able to estimate the LTV for each loan each month. I controlled for the Home Affordable Refinance Program (HARP) in the same context. HARP was implemented from 2009 to 2018 to help borrowers with little or no equity in their homes refinance into more affordable mortgages. Because I matched the refinanced (old) and refinanced (new) loans, I was able to observe whether the new loan was refinanced through HARP, allowing me to control for the policy factor that affects the equity distribution channel.

Another potentially important factor that could lead to refinancing inequality is the differential impact of lender-side credit supply tightness across income groups. Just as credit supply expanded disproportionately for low-income households during periods of credit expansion ([Mian and Sufi, 2009](#)), tightening credit supply could differentially affect low-income borrowers. To address this issue, I include two control variables. First, quarterly changes in the credit tightness index from the Federal Reserve’s Senior Loan Officer Survey were used. Second, I controlled for loan application denial rates by income quintile derived from Home Mortgage Disclosure Act (HMDA) data. Using the same income quintile boundaries that were used in the main loan panel, HMDA applications were divided into five income groups, and I used annual changes in loan application denial rates of each income quintile.

The baseline result of this regression using the simple rule of threshold ($\theta_{it} = 0$) is reported in [Table 1](#). Column (a) shows the unconditional probability, indicating that the highest income quintile has a 1.45 percentage points higher monthly refinancing probability compared to the bottom quintile. As control variables are gradually added, the final base model in column (4) shows a 1.15 percentage points difference.

The predicted refinancing probability by income quintile, calculated using the same regression results, is shown in [Figure 2](#). The predicted probability illustrates

Table 1: Average Refinancing Rate by Income Quintile (Simple Rate Gap)

Dependent Equation	Refinancing Indicator $\times 100$			
	(1)	(2)	(3)	(4)
Income Quintile 2	0.213 ^a (0.133)	0.316 ^a (0.115)	0.324 ^a (0.117)	0.296 ^a (0.131)
Income Quintile 3	0.072 ^a (0.115)	0.070 ^a (0.117)	0.077 ^a (0.117)	0.080 ^a (0.131)
Income Quintile 4	0.809 ^a (0.115)	0.647 ^a (0.117)	0.649 ^a (0.117)	0.470 ^a (0.131)
Income Quintile 5	1.452 ^a (0.133)	1.573 ^a (0.115)	1.672 ^a (0.117)	1.153 ^a (0.131)
Constant	1.923 ^a (0.064)	0.421 ^a (0.199)	0.504 ^a (0.218)	0.562 ^a (0.215)
<i>Control variables</i>				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,423	1,342,423	1,342,423	1,342,423
R-squared	0.002	0.010	0.011	0.011

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

Table 2: Average Refinancing Rate by Income Quintile (ADL gap)

Dependent Equation	Refinancing Indicator $\times 100$			
	(1)	(2)	(3)	(4)
Income Quintile 2	0.518 ^a (0.065)	0.562 ^a (0.068)	0.577 ^a (0.075)	0.579 ^a (0.072)
Income Quintile 3	0.924 ^a (0.093)	1.030 ^a (0.107)	1.064 ^a (0.113)	0.923 ^a (0.100)
Income Quintile 4	1.321 ^a (0.120)	1.472 ^a (0.125)	1.501 ^a (0.131)	1.148 ^a (0.118)
Income Quintile 5	1.872 ^a (0.139)	1.909 ^a (0.140)	2.057 ^a (0.151)	1.571 ^a (0.135)
Constant	1.940 ^a (0.064)	1.199 ^a (0.199)	1.317 ^a (0.218)	1.05 ^a (0.215)
<i>Control variables</i>				
LTV, credit, loan age, rate inc., ZIP		x	x	x
Credit tightness, HARP			x	x
Loan balance, Est. LTV				x
Observations	1,342,278	1,342,278	1,342,278	1,342,278
R-squared	0.001	0.013	0.014	0.014

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. Standard errors are clustered by zipcode and original vintage year.

the impact of differences in income quintiles on the final dependent variable when all control variables are at their mean values. Although there are some variations across quintiles with the addition of control variables, it is evident that the predicted probability increases clearly with income.

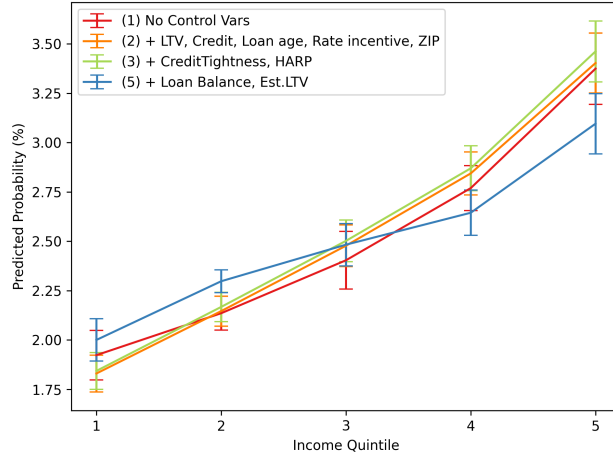


Figure 2: Predicted Refinancing Probability by Income Quintile

Notes: This plot shows the predicted monthly refinancing probability conditional on being in-the-money, using Equation 2. The 95% confidence intervals are represented by error bars and each line corresponds to the columns of the regression result in Table 1. Standard errors are clustered by ZIP code and original vintage year.

As shown in Table 2, the same regression is estimated using the *ADL gap*, which tightens the definition of being *in the money*. As previously discussed, this approach identifies refinancing profitability in a more conservative manner, requiring a larger gap between the coupon rate and the prevailing market rate compared with the simple threshold rule. Consequently, the estimated refinancing probabilities are higher than those in Table 1. Even under this stricter identification, high-income households exhibit significantly higher refinancing propensities than low-income households, with all coefficients statistically significant at the 1% level. Overall, after controlling for all covariates, households in the top income quintile show monthly refinancing probabilities of around 1.7-2.5%, while the bottom-quintile households refinance at least 50% less frequently, with a gap of roughly 1-1.1 % points.

3.2 Refinancing Probability by Rate Gap

To further substantiate the average refinancing probabilities, I decompose the observations by interest rate gap and compare refinancing activity across income groups.

As discussed in [subsection 2.4](#), I identify in-the-money in two ways: loans based on the rate gap between each loan’s current coupon rate and the prevailing market rate on newly originated loans with comparable characteristics (ZIP code, credit score, LTV, and income). In the baseline specification, I apply a simple rule that sets the refinancing threshold to zero, effectively assuming no upfront costs. As a robustness check, I also adopt the [Agarwal et al. \(2013\)](#) optimal refinancing rule, which endogenizes the threshold by incorporating closing costs, loan maturity, expected prepayment behavior, and interest rate volatility. This ADL-based threshold provides a more realistic measure of refinancing gaps while preserving comparability with the baseline setup.

$$RefiIndicator_{it} = \alpha + \sum_{Quintile \in Q} \sum_{Bin \in K} \beta_q^k \cdot \mathbf{1}[RateGap_{it} \in k] \cdot \mathbf{1}[IncomeQuintile_i \in Q] \quad (3)$$

$$+ \Gamma' X_{it} + \nu_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \quad (4)$$

I controlled the same variables as the previous regression using a non-parametric regression as specified in [Equation 3](#). Again, $Refi_Indicator_{it}$ is a indicator variable equal to 1 when the loan i is refinanced in month t . k denotes rate gap bins and j is the five income quintile at the time of origination. $\mathbf{1}[RateIncentive_{it} \in k]$ is a dummy variable equal to 1 when the rate gap falls in bin k , and X_{it} is a series of control variables. $\nu_{z(i)}$ and $\lambda_{v(i)}$ are ZIP code and vintage year-quarter fixed effects, respectively, and standard errors clustered by zip code and original vintage year.

The left panel of [Figure 3](#) presents the refinancing probability estimated using the simplest identification rule, where the threshold is set to zero. As shown, in the out-of-the-money region, refinancing rates are broadly similar across income quintiles. However, once loans enter the in-the-money region, a clear heterogeneity emerges. Even after accounting for the confidence bands of the first and fifth quintiles, high-income households are significantly more likely to refinance than low-income ones.

The right panel applies a more robust identification method based on the *ADL gap*, which allows the refinancing threshold to vary by loan and time (i, t). It is worth noting that this plot does not simply represent a horizontal shift of the left panel; rather, the bin assignment changes dynamically for each observation as the threshold evolves. This conservative specification already reveals refinancing inequality beginning around minus one percentage point, and the gap widens further in the in-the-money region.

Overall, regardless of the specific identification method, the pattern of refinancing inequality is consistently observed across nearly all rate-gap bins.

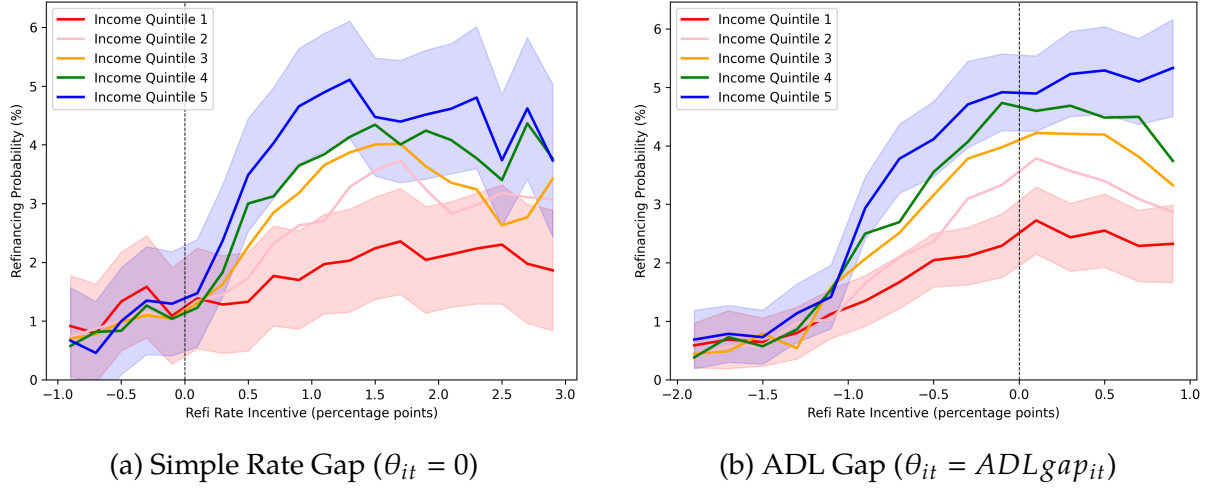


Figure 3: Refinancing Inequality by Rate Gap

3.3 Delays in Refinancing Decisions

In the previous regression analysis, I focused on the *intensity* dimension of refinancing inequality—how much borrowers refinance when it is financially beneficial to do so. Most of the existing literature on heterogeneous refinancing behavior has centered on this margin. However, because refinancing is an option-like decision, when borrowers choose to exercise the option is an equally important source of heterogeneity. The *timing* of refinancing may generate meaningful differences in realized savings even among borrowers with similar incentives.

To examine this timing margin, I align all loans at the month in which they first become in-the-money and compute, for each income quintile, the share of loans that remain outstanding over time. This approach yields a monthly survival rate of outstanding mortgages relative to the in-the-money point.

To implement this approach cleanly, I account for an important data feature. A complication arises because a loan may fall out-of-the-money shortly after first becoming in-the-money. In such cases, I treat these in-the-money occurrences as part of the same episode as long as the time gap between consecutive in-the-money months is less than twelve months. For example, suppose a loan becomes in the money and begins an episode, remains in that state for several months, then briefly falls out of the money, and becomes in the money again six months later. I treat this entire sequence

as a single in-the-money episode, since the interval between the two in-the-money observations is less than twelve months.

In this analysis, I use a more robust measure of being in-the-money based on the ADL gap, as described in [subsection 2.4](#). This measure is particularly important for analyzing refinancing timing, because the incentive to refinance declines sharply as a mortgage approaches maturity. The ADL-based measure incorporates this feature and therefore provides a more accurate basis for evaluating refinancing behavior over time.

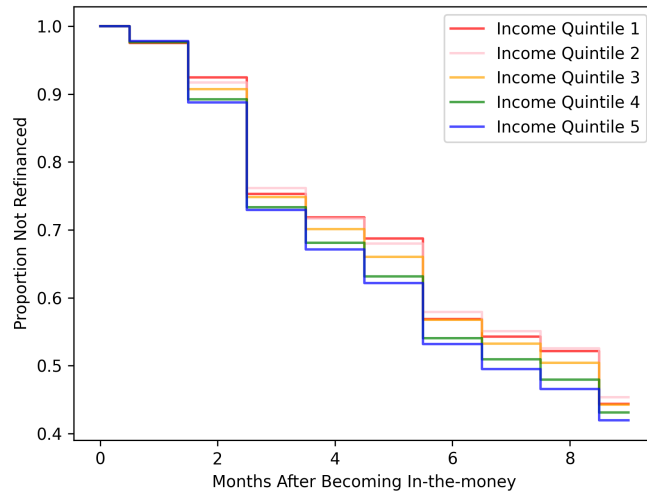


Figure 4: Proportion of Remaining Loans by Months After Becoming In-the-money

Figure 4 presents the resulting survival curves. Starting about two months after entering the money, loans held by households in the top income quintiles exhibit noticeably lower survival rates—meaning they refinance more quickly—than those in the bottom quintiles. This timing gap is persistent and economically meaningful. For example, the top quintile reaches a 50 percent survival rate after roughly seven months, whereas the bottom quintile takes about nine months. At the 40 percent threshold, the difference is even larger: ten months versus fifteen months. This analysis highlights a dimension of refinancing behavior that has received relatively little attention in the literature. To my knowledge, this is the first empirical evidence documenting a significant refinancing delay among lower-income households, complementing the existing focus on differences in refinancing intensity.

3.4 Robustness Checks

One of the major concerns is that since income quintiles are observed at the time of origination, changes in income quintiles at the time of refinancing might cause bias in regression results. Given that income quintile is a key variable, I performed four different robustness checks to validate the previous results.

First, I calculated the average loan age up to the refinancing point and used the Panel Study of Income Dynamics (PSID) data to track changes in mortgagors' income quintiles over the same period. In my sample, the average time to refinancing was 23 months. Therefore, I examined the changes in mortgagors' income quintiles over two years using the same yearly-changing income quintile thresholds. The probability of the lowest income quintile rising to the highest quintile after two years was 5.6%, while the probability of the highest quintile falling to the lowest quintile was 2.4%, both very low as suggested in [Table B.9](#)¹⁰. Although the data's limitations prevent tracking income changes, these results suggest that my estimates are unlikely to be significantly distorted by frequent changes in income quintiles.

Second, my data panel matches old loans with new loans, allowing for the observation of income and other characteristics at the time of refinancing. Therefore, I was able to perform the same regression using [Equation 2](#) with income quintile and other control variables at the time of refinancing. The results confirmed that the pattern consistent with the benchmark regression was maintained. As shown in [Figure B.7a](#), even with an increase in control variables, the predicted refinancing probability for high-income households remained significantly higher than for lower income quintile groups¹¹.

Third, I conducted a regression analysis on only those loans where there was no change in income quintile between the new and old loans. The results still showed a clear refinancing inequality, as illustrated in [Figure B.7b](#). This indicates that the previous results hold even when excluding changes in income quintile.

Lastly, since the probability of changes in a borrower's income quintile is lower for loans with shorter ages, I conducted repeated regression analyses by varying the

¹⁰Broadening the scope, the probability of the bottom two (1st and 2nd) income quintiles remaining in the same two quintiles after two years was 72.7%, whereas the probability of changing to the top two (4th and 5th) quintiles was 12.6%. Conversely, the probability of the top two income quintiles remaining in the same quintiles was 81.6%, while the probability of falling to the bottom two quintiles was 8.2%, indicating a low likelihood.

¹¹Note that this result is estimated based only on matched loans, excluding loans that were never refinanced. Therefore, the monthly refinancing probability is higher than in the benchmark regression. Also, the HARP variable, when based on new loans, showed that all loans were concentrated in quintiles 1 and 2. Controlling for this variable eliminated the differences between income quintiles. Therefore, it was not possible to control for the HARP variable.

age of the sample loans. The results showed a clear refinancing inequality regardless of the loan age cut. [Figure B.8](#) illustrates the predicted probabilities from regression (5), which includes all control variables, analyzed by progressively widening the loan age range: less than one, two, three, five, and seven years. As the loan age sample increased from one year or less to two years or less, there was an overall increase in probability, causing a line shift. However, it was still evident that higher income was significantly associated with higher refinancing probabilities. The same pattern persisted when increasing the loan age limit further.

Other robustness checks using the Survey of Consumer Finance (SCF) can be found in the Appendix [subsection B.2](#).

3.5 Potential Gains from Refinancing

Having identified significant refinancing inequality across income groups, I now investigate the extent to which this results in missed increases in disposable income. This study aims to examine how refinancing inequality ultimately impacts the monetary policy transmission channel, making the quantification of the saving amount crucial. I conducted the analysis on how much households' disposable income could have increased conditional on refinancing.

To start this analysis, I switched to a matched pair dataset of old and new loans. I analyzed how much each income group could reduce their interest rates through refinancing using a linear probability model. Using [Equation 2](#), I changed the dependent variable as the difference between the interest rate of the new loan and that of the old loan, leading to the specification of [Equation 5](#). I restrict the sample to in-the-money observations, since the calculation of missing gains is only meaningful when refinancing would have generated a financial benefit.

$$OldRate_i - NewRate_i = \alpha + \sum_{q=2}^5 \beta_q \cdot IncomeQuintile_{ij} + \Gamma' \cdot X_{it} + v_{z(i)} + \lambda_{v(i)} + \epsilon_{it} \quad (5)$$

The new interest rate after refinancing is generally lower than the existing one, negative beta values are expected. Since lower income quintiles typically have lower loan balances, a larger rate reduction is needed to offset the closing costs. Therefore, a larger absolute value of rate reduction is expected for lower income quintiles. As shown in [Figure A.1](#), the uncontrolled rate reductions of low quintile households are

significantly larger than that of high-income groups, but the gap diminishes as more control variables are added.

Table 3: Estimated Potential Savings from Refinancing

Income Quintile	Income (\$)	Avg. Loan Balance at Refi (\$)	Est. Rate Reduction by Refi (%p)	Mthly. Pymt. Reduction by Refi (\$)	Mthly. Pymt. Reduction to Income (%)
1	16,743	90,713	2.11	116.62	8.4
2	27,577	142,929	2.11	181.11	7.9
3	38,355	190,629	2.08	237.93	7.4
4	52,750	250,392	2.06	309.37	7.0
5	95,746	313,291	2.07	389.20	4.9

Notes: Assumed refinancing to 30-year FRM. Est. interest rate with a full set of control variables (eq.5). Sample restricted to in-the-money observations.

Based on the estimated rate reductions, I performed a back-of-the-envelope calculation to estimate the potential savings. By subtracting the rate reduction attributed solely to income quintiles from the existing interest rates, I derived the new post-refinancing interest rates. Assuming all refinancings are done with a 30-year fixed-rate mortgage, I calculated the monthly payment reduction using these new estimated rates and examined this as a proportion of income.

The results Table 3 shows that the bottom two income quintiles could have saved 7.9-8.4% of their monthly income through refinancing. This is a significant amount, as it represents additional disposable income occurring every month, not just a one-time saving. Although the absolute dollar amount of missed gains can be larger for higher-income borrowers—primarily because they tend to hold larger mortgage balances—the foregone savings constitute a much larger share of income for lower-income mortgagors. Given their higher marginal propensity to consume, the missing gains for these households likely translate into substantial unrealized consumption responses. Together, these findings suggest evidence that refinancing inequality could materially weakens the pass-through of monetary policy to disposable income and consumption.

4 Refinancing Response to Monetary Policy

The ultimate goal of this study is to examine whether the observed pattern of refinancing inequality impacts the monetary policy transmission mechanism. While the results so far have shown the *average* refinancing patterns across income groups over the period, it is necessary to assess the *dynamic* refinancing responses to changes in monetary policy to directly connect this pattern to monetary policy.

4.1 Data

To demonstrate refinancing inequality in a dynamic context, I examined the refinancing responses to exogenous monetary shocks across income quintiles. To do this, I followed the methodology of [Eichenbaum et al. \(2022\)](#) and used Freddie Mac panel data converted into regional data. The advantage of using regional aggregate data is twofold: first, it allows for the comparison and verification of results against established research; more importantly, it enables the use of the entire Freddie Mac population data rather than being limited to the old-new refinancing matched panel used in the previous analysis¹². While the earlier panel used loan-level data to meticulously control for loan and borrower characteristics, this analysis focuses on refinancing responses in a macroeconomic context, making regional aggregate data appropriate. I constructed a regional panel using Freddie Mac data, consisting of quarterly numbers of outstanding and refinancing loans by income quintile for mortgages originated from 2000 to 2019 across 908 three-digit ZIP code areas¹³.

For monetary policy shocks, I used the high-frequency shock data from [Swanson \(2021\)](#) to ensure exogeneity. Swanson recorded the changes in various asset prices within a 30-minute window around every FOMC meeting from 1991 to 2019. He then extracted the three largest principal components and demonstrated that these components correspond to changes in the federal funds rate, forward guidance, and large-scale asset purchases (LSAPs). A key advantage of this approach is that it addresses the limitations of high-frequency identification during the zero lower bound (ZLB) period. Since a significant portion of my data includes the ZLB period, Swanson's shocks are particularly relevant.

I obtain aggregate time-series variables, including forecasts of unemployment, inflation, and GDP, from the Survey of Professional Forecasters (SPF). These variables were used to control for the information channel suggested by [Nakamura and Steinsson \(2018\)](#). Also, to control for house prices, I used the Zillow ZIP code-level house price index.

¹²Although refinancing information (reasons for prepayment) is not available at the loan level in the Freddie Mac data, which required me to match old and new loans to ensure that the loan was prepaid due to refinancing, it can be observed in the origination data. By aggregating the issuance data by region, I can use the entire dataset to derive the share of refinances relative to total outstanding loans.

¹³However, the actual analysis used data from 2004 onwards. This is because Freddie Mac data provides information only on loans originated from 2000 onwards, leading to insufficient cumulative loan data by income quintile up to 2003. Therefore, to prevent outlier values in income quintile refinancing rates, a sort of burn-in period was necessary.

In this analysis, the key variable is the refinancing propensity. I compute this by measuring the cumulative proportion of refinanced loans from outstanding loans at the regional level as below:

$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q} + \dots + x_{t+h}^{r,q}}{(y_t^{r,q} + \dots + y_{t+h}^{r,q})/h} \quad (6)$$

Here, $x_t^{r,q}$ is the number of refinancing loans by income quintile q in region r in quarter t , $y_t^{r,q}$ is the number of all outstanding loans, and h is the analysis time horizon.

One limitation is worth noting before proceeding. Although the underlying loan-level data cover originations from 2000 onward, I begin the regression sample in the second quarter of 2004 to allow for a sufficient burn-in period—that is, until the stock of outstanding loans becomes large enough. Given that the identified monetary policy shocks exists up to the second quarter of 2019 and that the regressions include lagged variables, the effective time dimension consists of 56 quarterly observations. This relatively short time span makes clustering by time infeasible, as it would result in excessively large standard errors. At the same time, clustering at the monthly level is infeasible since the matched-sample structure does not provide sufficient observations per month. To address this small- T issue, I instead employ block bootstrapping by time, which corrects the standard errors for the limited number of time periods. In addition, following the recommendation of Almuzar and Sancibrian (2024), I include a sufficient number of lags (six quarters) to mitigate this concern.

4.2 Regression Results

Average Refinancing Response to Monetary Policy Shocks I begin by examining the average refinancing response for all income groups, in comparison with existing research. I start by considering the local projection analysis suggested by Jordà (2005).

$$\rho_{t+h}^r = \beta_0^h X^r + \beta_1^h \Delta R_t^{IV} + \beta_2^h Z_t^r + \beta_3^h Z_t + \text{Lags}_t^r + \epsilon_t^z \quad (7)$$

Here, X^r represents a vector of region fixed effects, and ΔR_t^{IV} denotes an instrument variable for interest rate *falls* using using high-frequency shocks. The interest rate is transformed to a positive sign indicating a decrease in interest rates. Therefore, I expect a positive response of refinancing. Z_t^r includes regional control variables such

as changes in unemployment rates and house prices. The vector Z_t denotes a set of time-varying controls, specifically including a two-year ahead horizon, the civilian unemployment rate (also two years ahead), and the CPI inflation rate (forecasted for both one and two years ahead). I used three lags terms for all variables.

First, I show the analysis by demonstrating the strong impact of monetary policy on mortgage interest rates. [Table 4](#) presents the first-stage estimates. All three high-frequency factors exhibit a significant positive correlation with the 30-year fixed mortgage interest rate. In particular, the effect of LSAP, which involves the purchase of long-term bonds, has the largest impact on the 30-year fixed mortgage rate, aligning with expectations. The negative relationship between rising expectations for GDP and CPI and the actual mortgage rate reduction effect is consistent with economic interpretation, as higher expectations for GDP and CPI reduce the real impact of mortgage rate cuts. The F test for the joint significance of the regression coefficients is greater than ten.

Table 4: First-stage Estimates

Dependent		Δ Real Mortgage Rates of 30-year FRM	
High-frequency Factors		Survey of Professional Forecasters	
- FFR	0.0165 ^a (0.00)	- Δ GDP (2 yrs)	-0.0341 ^a (0.00)
- Forward Guidance	0.0290 ^a (0.00)	- Unemp. (2 yrs)	0.0273 ^a (0.00)
- LSAP	0.0344 ^a (0.00)	- CPI (1 yrs)	-0.0347 ^a (0.00)
Δ House Price	0.0005 ^a (0.00)	- CPI (2 yrs)	-0.0832 ^a (0.01)
Constant	0.0427 ^a (0.02)	Unemp.	0.00641 ^a (0.00)
Observations	46,351	F-statistic	2,803.820
R-squared	0.358	Prob. > F	0.000

Notes: Superscript a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis.

[Figure A.3](#) shows the average refinancing response to a 1% interest rate falls. It indicates that cumulatively 15% of all mortgages are refinanced 7 quarters after the shock. After the seventh quarter, the cumulative ratio decreases because new mortgages increase the total number of outstanding loans while the number of refinancing decreases, gradually lowering the ratio below the unconditional ratio. Although the types of mortgages covered by the data and the sample periods differ from [Eichenbaum et al. \(2022\)](#)'s results, which show about 15-25%¹⁴ refinancing by the fifth quarter following an interest rate shock, the result is consistent with their findings.

¹⁴They estimated the refinancing rate depending on the interest rate gaps between existing loans and the market rate, which makes the estimated range larger.

Heterogeneous Refinancing Response Across Income Groups Now, I will examine the heterogeneous refinancing responses across different income levels, which is the primary focus of this chapter. By modifying the Equation 7 I consider following specification:

$$\rho_{t+h}^{r,q} = \beta_0^h X^r + \beta_1^h \Delta R_t^{IV} + \beta_2^h Z_t^r + \beta_3^h Z_t + Lag s_t^{r,q} + \epsilon_t^z \quad (8)$$

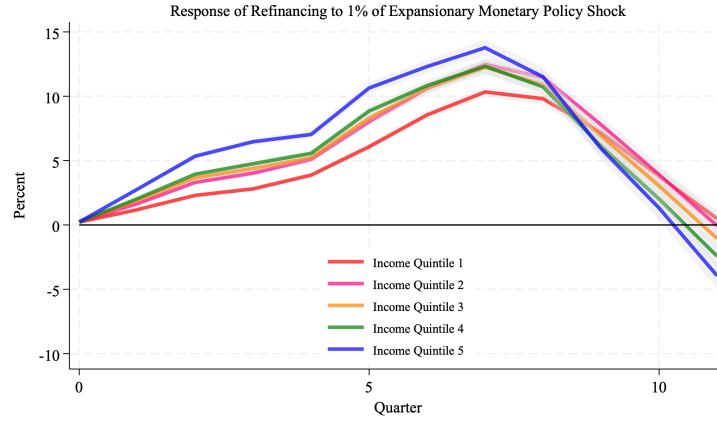


Figure 5: Cumulative Refinancing Response to a 1% Interest Rate Cut by Income Quintile

Notes: This plot shows the cumulative refinancing response by income quintile to 1% interest rate falls using Equation 8. The dependent variable is the cumulative proportion of loans refinanced in each region and income group as described in Equation 6. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to 2022 was used. The shading represents the 95% confidence interval.

While the meaning of each variable remains the same as in Equation 7, the dependent variable, which is the proportion of refinanced loans, was analyzed separately for each income level. Figure 5 illustrates the refinancing response by income quintiles to a 1% interest rate reduction. Seven quarters after the interest rate reduction, households in the highest income quintile show a cumulative refinancing rate of approximately 15%. In contrast, households in the bottom income quintiles have peak cumulative refinancing rates of about 10%, significantly lower than their wealthier counterparts.

There are other specifications to measure the dependent variable of the refinancing proportion. As explained in the Appendix subsection B.3, I conducted robustness checks using different specifications and the pattern of refinancing inequality was robust across all specifications.

5 A Model for Heterogeneous Refinancing Friction

To quantitatively assess how the refinancing inequality patterns documented in the empirical analysis affect the transmission of monetary policy, this paper develops a structural model of household mortgage refinancing. The model builds on the framework of [Berger et al. \(2021\)](#), but departs from the baseline specification in an important dimension. In the original model, refinancing inequality does not emerge: instead, lower-income households—due to their limited liquid asset holdings—behave as wealthy hand-to-mouth agents and participate more actively in refinancing, generating a pattern opposite to that observed in the data. By incorporating three additional mechanisms described below, the proposed model successfully reproduces the empirically documented refinancing inequality, whereby higher-income households refinance more frequently while lower-income households exhibit systematically lower refinancing intensity.

The model improves upon the existing framework along three key dimensions. First, while the baseline model relies solely on an inattention channel to rationalize low refinancing rates, this paper distinguishes between two conceptually different sources of refinancing frictions: inattention and hassle costs. Separating these frictions allows the model to capture distinct behavioral and economic mechanisms underlying refinancing decisions. Second, the baseline specification models attention as a simple on-off switch depending on whether a household is in the money, which fails to capture how attention varies with the magnitude of the refinancing incentive. This limitation is quantitatively important, as the strength of monetary policy transmission may depend on the size of interest rate cuts. To address this, the model introduces a gap-dependent attention intensity rather than a binary attention mechanism. Third, and most importantly, the model allows both inattention and hassle cost frictions to vary by income state. Attention heterogeneity is disciplined using a parsimonious parametric structure consistent with empirical patterns observed in the data, while refinancing thresholds associated with hassle costs are estimated fully nonparametrically.

5.1 Model setup

The economy is populated by a continuum of risk-averse households who solve an infinite-horizon optimization problem in continuous time. Households face both aggregate interest rate risk and idiosyncratic labor income shocks. The aggregate short rate follows an exogenous process and is controlled by the monetary authority.

Households are heterogeneous with respect to labor assets, outstanding mortgage balances, coupon rates, and income. Each household holds a long-term fixed-rate mortgage and can choose whether to refinance, including the option to extract home equity through cash-out refinancing. Refinancing requires paying a fixed cost and, in addition, households face two state-dependent refinancing frictions that are described below. On the supply side, lenders are risk-neutral. Taking the exogenous short rate as given, lenders price mortgage contracts based on expected prepayment behavior summarized by endogenous hazard rates, and supply mortgages such that equilibrium mortgage rates clear the mortgage market

5.2 Mortgage Refinancing and Frictions

Mortgage refinancing in the model is governed by two distinct and complementary frictions, which I explicitly *decompose* to isolate their separate economic roles. The first is a *threshold-type friction*, capturing the fact that refinancing entails not only explicit monetary fixed costs but also non-monetary costs, such as time costs, information-processing costs, and search costs for better mortgage terms. If this friction were the only impediment to refinancing, households would refinance with probability one whenever the gains from refinancing exceed a sufficiently high threshold.

The second friction is *inattention*. Even when refinancing gains are large enough to overcome threshold costs, observed refinancing rates remain far below one. This pattern suggests that households do not continuously monitor refinancing opportunities, but instead pay attention only intermittently. The model captures this behavior through a Calvo-style attention friction.

By decomposing refinancing frictions into an execution margin (threshold) and an attention margin (frequency), the model allows us to separately quantify the contribution of each channel to refinancing activity and to conduct counterfactual exercises that shut down individual frictions. This decomposition is central to understanding how refinancing inequality shapes the transmission of monetary policy.

Hassle-Cost Channel The threshold-type friction is interpreted as a hassle-cost (execution) channel, reflecting the costs associated with executing a refinancing transaction. Information-processing and search costs depend on financial sophistication and education, while the time required to evaluate and complete refinancing entails an opportunity cost that increases with income. As a result, the effective hurdle to refinancing may vary systematically across income groups.

Formally, the hassle-cost channel is modeled as an income-dependent refinancing threshold. A household executes a refinancing transaction only if the value gain from refinancing exceeds an income-specific threshold. To avoid imposing restrictive functional form assumptions, these thresholds are estimated fully nonparametrically across income groups. The resulting execution rule is summarized in equation (9), which characterizes refinancing as a comparison between the value gain from refinancing and the income-dependent hassle-cost threshold.

$$\rho^{refi} = \begin{cases} 1, & \text{if } V^{\text{refi}} - V^{\text{stay}} > \Phi_s \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Attention Friction The attention channel governs how frequently households pay attention to refinancing opportunities over calendar time. Existing models typically assume that attention switches discretely on or off depending on whether a household is in the money. However, this binary formulation fails to capture a key behavioral feature of refinancing behavior: households naturally become more attentive to refinancing as the gap between their current mortgage coupon rate and the prevailing market rate widens. This distinction is quantitatively important, as the strength of monetary policy transmission may depend on the magnitude of interest rate movements rather than solely on whether refinancing is profitable.

In the model, attention varies continuously with the refinancing incentive, measured by the interest rate gap. Households receive opportunities to consider refinancing according to a Poisson process, whose arrival intensity depends on both the interest rate gap and the household's income state. Attention intensity increases as the household becomes deeper in-the-money.

The functional form of attention is disciplined by two robust empirical patterns observed in the data. First, even for households that are deep in the money, attention to refinancing remains limited rather than approaching one. Second, the maximum level of attention differs systematically across income groups. These features are captured by a parsimonious, gap-dependent attention specification, formalized in equation (10), which allows for income-dependent heterogeneity in maximum attention capacity.

$$\chi(gap_t, s_t) = \chi_{\max}(s_t) \cdot \frac{1}{1 + \exp(-k [gap_t - \mu])} \quad (10)$$

Here, $gap \equiv m_t^* - m_t$ denotes the difference between the household's current mortgage coupon rate (m_t^*) and the prevailing market mortgage rate (m_t). $\chi_{\max}(s_t)$ denotes the maximum attention capacity for households in income state s , capturing the upper bound on how frequently a household can pay attention to refinancing opportunities even when it is deep in the money. The parameter μ governs the location of the attention response, representing a region in which refinancing incentives are too small to meaningfully trigger attention. The slope parameter k controls how sensitively attention responds to changes in the interest rate gap, that is, how rapidly attention intensity increases as the gap widens. Since the mortgage coupon rate is fixed, movements in the interest rate gap are driven by changes in the market interest rate, so k captures how households' refinancing attention responds to monetary policy-induced declines in market rates.

5.3 Household Problem

Households maximize lifetime utility by choosing both their consumption (C_t) path, the mortgage refinancing (ρ_t^{refi}), and cash-out decision (ΔF_t^{co}). These decisions are made subject to the structure of the mortgage contract and the household's intertemporal budget constraint. Given an income process and the prevailing interest rate environment, each household dynamically decides how much to consume today and when to refinance its existing mortgage.

$$\begin{aligned}
V(S_t, r_t) &= \sup_{\{C_t, \rho_t^{refi}, \Delta F_t^{co}\}} \mathbb{E}_t \left[\int_0^\infty e^{-\delta t} \frac{C_t^{1-\gamma}}{1-\gamma} dt \right], \\
\text{s.t. } dW_t &= (Y_t - C_t + r_t W_t - m_t^* F_t) dt + \underbrace{dN_t^\chi \rho_t^{refi}}_{\text{refi decision}} \left(\underbrace{\Delta F_t^{co}}_{\text{cash-out}} - \underbrace{\kappa}_{\text{refi fixed cost}} \right), \\
dF_t &= \underbrace{-\alpha F_t dt}_{\text{amortization}} + \underbrace{(F'_t - F_t) dN_t^\chi \rho_t^{refi}}_{\text{refi jump}}, \\
dm_t^* &= \underbrace{(m_t - m_t^*) (dN_t^m + dN_t^\chi \rho_t^{refi})}_{\text{coupon reset at refinancing (moving + refi)}}.
\end{aligned}$$

The value function $V(S_t, r_t)$ represents a household's maximal expected lifetime utility given its current state. The individual state is $S_t \equiv (W_t, F_t, m_t^*, s_t)$, where W_t denotes liquid wealth, F_t the outstanding mortgage balance, m_t^* the existing mortgage

coupon rate, and s_t the idiosyncratic income state. The aggregate state of the economy is summarized by the short rate r_t . The vector χ collects the parameters governing refinancing frictions, including the attention and execution channels introduced above.

Households choose consumption C_t and refinancing-related actions, including whether to execute refinancing and the size of cash-out refinancing ΔF_t^{co} , to maximize discounted expected utility over an infinite horizon. Preferences are represented by a CRRA utility function with risk aversion parameter γ , and future utility is discounted at rate δ .

Refinancing opportunities arrive stochastically. In particular, households receive opportunities to consider refinancing according to a Poisson process dN_t^χ with intensity $\chi(gap_t, s_t)$, where the intensity depends on the refinancing incentive and the current income state. Conditional on such an arrival, households decide whether to execute refinancing subject to the execution (hassle-cost) threshold. In addition, exogenous events such as relocation or other forced mortgage resets trigger refinancing according to an independent Poisson process dN_t^m . These arrival processes jointly govern the timing of mortgage refinancing and coupon resets in the model.

It is important to emphasize that income affects refinancing behavior in a *non-monotonic* manner, even absent any mechanical targeting of refinancing frictions. On the one hand, lower-income households are more likely to be constrained borrowers. They face tighter payment-to-income (PTI) constraints and may lack the liquidity required to cover fixed refinancing costs. As a result, even when refinancing opportunities arise, these constraints limit their ability to act, giving rise to a *constraint channel* that lowers refinancing probabilities among low-income households.

On the other hand, lower-income households exhibit a strong *liquidity motive*. Because they place a higher value on immediate liquidity, they have stronger incentives to extract home equity through cash-out refinancing. Conditional on being able to refinance, they therefore tend to refinance more aggressively. The overall effect of income on refinancing and cash-out policy functions is thus non-monotonic, reflecting the interaction between these two opposing forces rather than a mechanical consequence of the model's frictional structure.

5.4 Uncertainty Environment

The model features two sources of uncertainty. First, households face idiosyncratic labor income risk, following the AR(1) process:

$$d \ln Y_t = \eta_y dt + \sigma_y dZ_t^y \quad (11)$$

Each household's labor income Y_i evolves stochastically according to a mean-reverting process with persistence parameter η_y and volatility σ_y . The innovation is driven by a standard Brownian motion Z_t^r that is independent across households and orthogonal to all aggregate shocks. This idiosyncratic income component introduces cross-sectional heterogeneity even in the absence of aggregate fluctuations.

Second, the aggregate short-term interest rate r_t follows a continuous-time stochastic process of the Cox–Ingersoll–Ross (CIR):

$$dr_t = -\eta_r(r_t - \bar{r})dt + \sigma_r\sqrt{r_t}dZ_t^r, \quad (12)$$

It is characterized by persistence η_r , volatility σ_r , and long-run mean $\bar{r}$. The process is driven by a common Brownian motion Z_t^r , which captures aggregate shocks affecting all households simultaneously. Changes in r_t propagate through to mortgage rates and refinancing incentives, thereby linking aggregate monetary conditions to household-level refinancing dynamics.

Together, the idiosyncratic income process (Y_i) and the aggregate short-rate process (r_t) generate both individual and aggregate uncertainty in the model. These shocks jointly shape the evolution of consumption, wealth accumulation, and mortgage refinancing behavior across the household distribution.

5.5 Mortgage Market Equilibrium

Each mortgage is modeled as a fixed-rate loan contract characterized by its coupon rate m and outstanding balance F . Instead of specifying a detailed amortization schedule, the model assumes that a constant fraction α of the loan is repaid each year. This simplification implies that repayments accelerate slightly as the loan approaches maturity, but it greatly reduces computational complexity without affecting the key results. When a household refinances, the coupon is replaced by a new rate m_t^* drawn from the prevailing market rate.

The mortgage price satisfies a risk-neutral pricing condition that accounts for the expected prepayment probability. Lenders fund themselves at the short rate and earn expected returns from mortgage payments such that, in equilibrium, they break even. In other words, the present value of a newly issued mortgage equals its face value, and the equilibrium coupon $m(r_t)$.

A key assumption is that lenders do not observe each household's refinancing friction. Instead, they form expectations based on the average refinancing hazard across the population $(\bar{\chi})$.¹⁵

$$P(W, F, r, m^*, s; \bar{\chi}) = \mathbb{E} \left[\int_0^\tau e^{-\int_0^t r_d du} m^* dt + e^{-\int_0^\tau r_d du} \mid r_0 = r \right] \quad (13)$$

Under the assumption of complete markets, the equilibrium mortgage rate is determined such that the lender's expected profit is zero after accounting for anticipated refinancing behavior.

$$P(W, F, r, m^*, s; \bar{\chi}) = 1 \quad (14)$$

5.6 Dynamic Programming and Distributional Dynamics

Hamilton–Jacobi–Bellman Equation Given the continuous-time formulation and the presence of stochastic refinancing opportunities, the household's optimization problem admits a Hamilton–Jacobi–Bellman (HJB) representation. Let $V(S, r)$ denote the value function, where the individual state is $S = (W, F, m^*, s)$ and the aggregate state is summarized by the short rate r .

The HJB equation is given by

$$\delta V(S, r) = \max_C \left\{ u(c) + V_W(S, r)(Y(s) - C + rW - m^*F) - \alpha F V_F(S, r) \right. \quad (15)$$

$$+ \mathcal{L}_r V(S, r) + \lambda^\chi(S, r) \left[\max\{V^{\text{refi}}(S, r), V(S, r)\} - V(S, r) \right] \quad (16)$$

$$\left. + \lambda^m \left[V^m(S, r) - V(S, r) \right] \right\}. \quad (17)$$

The first line captures the standard flow utility and endogenous savings decision.

Consumption C is chosen optimally subject to the household's intertemporal budget

¹⁵As noted by Berger et al. (2024), households with higher attention parameters are overrepresented among those who actually prepay, relative to the ex-ante distribution. This discrepancy between ex-ante and ex-post distributions can affect the lender's perceived hazard rate and, consequently, mortgage pricing. In the current version of the model, this feedback is not yet incorporated but will be addressed in a later refinement of the framework.

constraint, and V_W and V_F denote derivatives of the value function with respect to liquid wealth and mortgage balance, respectively. The operator $\mathcal{L}_r$ captures the effect of aggregate interest rate dynamics on the value function.

The second line reflects endogenous refinancing opportunities. Refinancing consideration arrives according to a Poisson process with intensity $\lambda^x(S, r) = \chi(\text{gap}, s)$, where $\text{gap} = m^* - m(r)$ denotes the refinancing incentive. Upon such an arrival, households compare the continuation value from refinancing, $V^{\text{refi}}(S, r)$, with the value of not refinancing. The execution decision is governed by the hassle-cost threshold, so refinancing occurs only when the value gain exceeds the income-state-dependent threshold $\Phi(s)$.

The final term captures exogenous refinancing events, such as relocation, which arrive with constant intensity λ^m and mechanically reset the mortgage coupon to the prevailing market rate.

Distribution Dynamics Let $g_t(S, r)$ denote the cross-sectional distribution of households over individual states S , conditional on the aggregate state r . The evolution of the distribution is governed by a Kolmogorov Forward Equation (KFE):

$$\begin{aligned} \frac{\partial g_t(S, r)}{\partial t} = & -\nabla_S \cdot (\mu(S, r) g_t(S, r)) \\ & + \lambda^x(S^-, r) \mathbf{1}\{\text{refi}\} g_t(S^-, r) - \lambda^x(S, r) \mathbf{1}\{\text{refi}\} g_t(S, r) \\ & + \lambda^m g_t(S^-, r) - \lambda^m g_t(S, r), \end{aligned}$$

where $\mu(S, r)$ collects the deterministic drifts implied by savings and mortgage amortization, and S^- denotes the pre-refinancing state. The first term captures the continuous drift in liquid wealth and mortgage balances induced by consumption, income, and amortization. The remaining terms reflect jump transitions associated with refinancing. Endogenous refinancing transitions occur with intensity $\lambda^x(S, r)$ and depend on the household's refinancing policy, while exogenous refinancing events arrive at rate λ^m . These jump transitions reallocate mass across the state space according to the optimal cash-out policy and coupon reset rules.

In the numerical implementation, the HJB and KFE are solved jointly using an implicit finite-difference scheme with upwinding for the drift terms and sparse transition matrices for the jump components.

6 Quantitative Analysis

6.1 Calibration

Attention Parameters The attention friction is governed by the parameters $\chi_{\max}(s)$, k , and μ . I discipline these parameters using a combination of micro-level survey evidence and reduced-form patterns in the refinancing data.

The parameter $\chi_{\max}(s)$ is interpreted as the maximum attention capacity of households in income state s . In the model, $\chi_{\max}(s)$ determines the upper bound of refinancing intensity when households are deep in the money. Empirically, this parameter maps directly to the ceiling of observed refinancing probabilities across income groups, a feature that is clearly visible in the data.

To provide a microfoundation for income-dependent heterogeneity in attention capacity, I use household-level proxy measures constructed from the National Survey of Mortgage Originations (NSMO). Specifically, I exploit survey questions that record (i) the number of lenders a borrower considered and (ii) the number of lenders to which the borrower applied when originating a mortgage. These measures capture the intensity of search and information acquisition during mortgage origination. I normalize the responses and aggregate them into a single attention index. The resulting index is then mapped to the model's income states using nonlinear interpolation to align survey income categories with the model's income groups. Details of the construction of this index and the interpolation procedure are provided in Appendix.

Conceptually, these survey-based measures proxy the intensive margin of attention—how much effort households exert when actively searching for mortgage terms—rather than the extensive margin of whether households pay attention at all. I adopt two identifying assumptions. First, mortgage origination is a setting in which households exert close to maximal feasible search effort, allowing us to interpret the survey-based index as an empirical proxy for maximum attention capacity. Second, I assume that income-dependent heterogeneity in maximum attention at origination persists into the refinancing stage. Under these assumptions, the cross-sectional variation in the survey-based attention index directly disciplines $\chi_{\max}(s)$.

Gap Sensitivity Parameters The remaining attention parameters, k and μ , govern how attention intensity responds to the refinancing incentive, measured by the interest rate gap. The parameter μ controls the location of the attention response, while k determines the sensitivity of attention intensity to changes in the gap. I restrict these parameters to be common across income states.

This restriction is motivated by reduced-form evidence from the data. Across income groups, the increase in refinancing activity as the interest rate gap widens exhibits similar curvature and timing, with differences across groups primarily reflected in the maximum refinancing rates rather than in the responsiveness to the gap itself. Allowing $\chi_{\max}(s)$ to vary by income state while keeping k and μ common is therefore sufficient to capture the observed heterogeneity in refinancing behavior.

I estimate k and μ using Nonlinear Least Squares (NLS). Specifically, for each income quintile, I fit the gap-dependent attention specification implied by equation (10) to the empirical refinancing hazard as a function of the interest rate gap. The estimation jointly disciplines the slope and location of the attention response while allowing the level to differ across income groups through $\chi_{\max}(s)$. This approach ensures that the calibrated attention function matches the observed shape of refinancing responses to interest rate incentives without overfitting income-specific sensitivities.

Other model parameters The other model parameters follows the quantitative macro-finance literature. The coefficient of relative risk aversion is set to $\gamma = 2$, a standard value in macroeconomic models. The baseline mortgage balance F , the time-preference parameter δ , and the parameters governing the interest rate process — $(\eta_r, \sigma_r, \bar{r})$ — are calibrated to match [Berger et al. \(2021\)](#), implying a mortgage balance of \$150,000, an annual mean short rate of 3.5%, and an interest-rate volatility of 6%. The stochastic income process follows [Floden and Lindé \(2001\)](#), with a log-income persistence corresponding to a half-life of 7.3 years and an annualized volatility of 21%.

For refinancing frictions, two types of behavioral parameters, $\chi_1(s)$ and $\chi_2(s)$, capture the heterogeneous intensity of refinancing attention and the execution delay across income quintiles, respectively. Both parameters are modeled as five-element vectors corresponding to income quintiles $s = 1, \dots, 5$.

The payment-to-income ratio $\phi^{PTI} = 0.43$ limits the maximum feasible mortgage payment relative to income, while the loan-to-value ratio $\phi^{LTV} = 0.8$ restricts the maximum outstanding balance relative to home value. Both parameters are calibrated to match average underwriting standards in the Freddie Mac.

Further details on the cross-sectional distribution of the baseline model are provided in [Figure A.5](#).

Table 5: Model Parameter Values

Panel A: Exogenous Processes			
Parameter	Value	Description	Source
$\ln 2/\eta_r$	5.3 years	Half-life of interest-rate shock	Berger et al. (2021)
$\bar{r}$	3.5% (p.a.)	Unconditional mean short rate	Berger et al. (2021)
σ_r	6% (p.a.)	Volatility of short rate	Berger et al. (2021)
$\ln 2/\eta_y$	7.3 years	Half-life of (log) income shock	Flodén and Lindé (2001)
$\mathbb{E}[Y_t]$	\$58,000	Unconditional mean income	Flodén and Lindé (2001)
σ_y	21% (p.a.)	Log-income volatility	Flodén and Lindé (2001)
γ	2	Inverse of IES	Macro literature standard
F	\$150,000	Mortgage balance outstanding	Berger et al. (2021)

Panel B: Refinancing Frictions			
Parameter	Hybrid	Description	Source
ν	4.1% (p.a.)	Arrival rate of exogenous moving shocks	Berger et al. (2021)
$\chi_{\max}(s)$	{0.020, 0.023, 0.227, 0.202, 0.212}	Maximum attention capacity by income quintile	Estimated (NSMO)
k	1.931	Responsiveness of attention to rate gap	Estimated (Freddie Mac)
μ	0.284	Location parameter of attention response to rate gap	Estimated (Freddie Mac)
κ	\$3,000	Fixed (menu) cost of refinancing	Eichenbaum et al. (2021)
ϕ^{PTI}	0.43	Payment-to-income constraint ratio	Freddie Mac
ϕ^{LTV}	0.80	Maximum loan-to-value ratio	Freddie Mac

6.2 Results: Hazard Rate

I now turn to evaluating the performance of the model. As a first step, I examine how well it replicates the targeted moments—refinancing hazard rates by rate-gap bins across income quintiles. The exercise is conducted by feeding the market coupon distribution and the realized short-rate path as of January 2005 into the model¹⁶, and comparing the model-generated outcomes with the empirical estimates from the data.

The goal is to assess how well the model reproduces the heterogeneous refinancing patterns documented in the data. In particular, the calibration targets two key dimensions of refinancing behavior: *intensity* and *timing*. Refinancing intensity captures how sensitively households respond once they become in-the-money—driven mainly by the attention parameter, χ_1 . Refinancing timing captures how quickly households act after recognizing the benefit. Even when refinancing is clearly optimal, many households delay execution—a friction captured by χ_2 .

Figure 6 shows the model fit against the data. Although the model is not yet fully calibrated, it replicates several salient empirical patterns. The baseline model (blue dashed) reproduces the higher refinancing peaks observed among the fourth and fifth income quintiles compared to the bottom groups.¹⁷ By contrast, when income-specific heterogeneity is suppressed and homogeneous frictions are imposed, the

¹⁶The empirical sample begins in 2000, but the model simulation starts from the mid-sample to allow for a sufficient burn-in period.

¹⁷The current calibration is preliminary and will be refined further.

model (orange dotted) tends to overpredict refinancing among low-income borrowers and underpredict it among high-income borrowers, generating systematic distortions relative to the data.

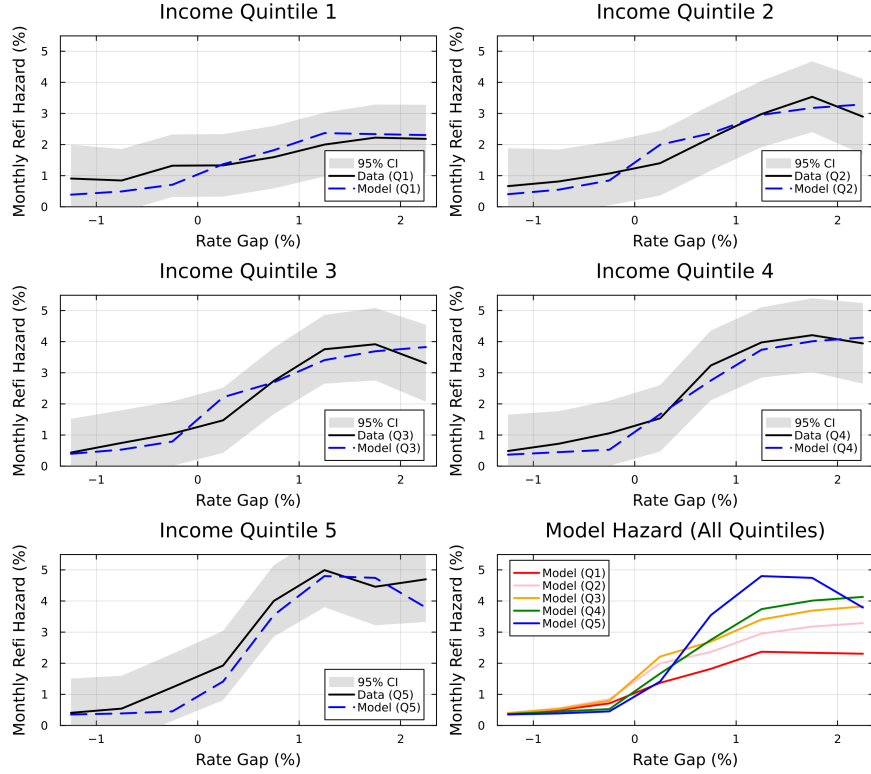


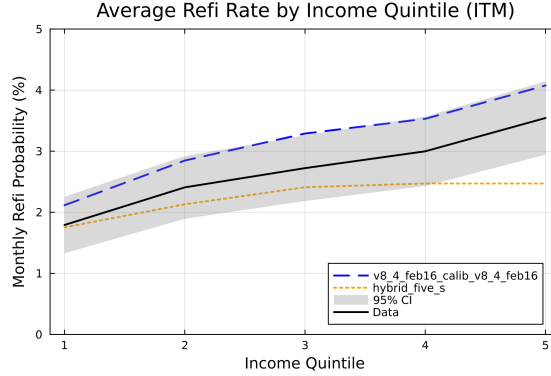
Figure 6: Hazard Rate by Gap Bin: Model vs. Data

I also compare the average refinancing rate among *in-the-money* borrowers across income quintiles. Figure 7 Panel (a) shows that while the baseline heterogeneous model produces somewhat higher overall refinancing levels than in the data, it can capture the upward-sloping pattern of refinancing inequality, refinancing rates increasing with income. In contrast, the homogeneous χ model generates an almost flat profile across income quintiles, failing to match the observed heterogeneity in refinancing behavior.¹⁸

Figure 7 Panel (b) compares this baseline model result with the pure Calvo-type model of Berger et al. (2021), which excludes both the menu cost and the cash-out margin. Their baseline sample period and data coverage differ from mine, leading to somewhat lower overall refinancing rates. It is important to note that their model produces identical refinancing probabilities across all income quintiles.

¹⁸Even under the homogeneous- χ specification, the model still produces slight variation in refinancing probabilities across income groups. This arises because the presence of the cash-out channel and the menu cost in my model continues to link refinancing incentives to liquidity and income levels.

As a result, once a single friction is imposed uniformly across income groups, the model cannot reproduce the observed pattern of refinancing inequality in data and thus cannot capture the heterogeneous transmission of monetary policy.



(a) Baseline vs Berger et al. (2021)

Figure 7: Average Hazard Rate by Income Quintile: Model vs. Data

6.3 Results: Consumption Response to Monetary Policy

The ultimate goal of this paper is to assess how refinancing inequality affects the transmission of monetary policy. When interest rates fall, some borrowers become *in the money*, yet frictions may prevent them from refinancing. If refinancing is delayed or forgone, the expansionary effects of monetary policy are weakened.

To quantify this mechanism, I compare the consumption response to a 100-basis-point decline in the policy rate across different frictional environments. The main experiment relaxes the in-the-money attention friction (χ_{max}) while holding other parameters fixed. This counterfactual is realistic and policy-relevant: attention frictions can be mitigated in practice, as even simple reminders have been shown to meaningfully increase refinancing activity.¹⁹

Figure 8 illustrates the consumption responses to a 100 basis-point interest rate cut under the baseline and the counterfactual scenario with higher χ_{max} . In the counterfactual, all income groups are assumed to have the same level of attention as the top quintile in the baseline model. As shown in the left panel, the counterfactual produces a stronger and more persistent consumption response starting immediately after the policy shock. The right panel plots the cumulative difference between the

¹⁹See Byrne et al. (2023) for evidence from a randomized controlled trial. In contrast, the execution friction Φ_s represents deeper behavioral and structural costs—such as time costs, paperwork burden, or limited financial literacy—which are more difficult to alleviate. Therefore, focusing on the χ_{max} channel highlights how small improvements in attention can generate disproportionately large policy effects.

two paths. By two and a half years after the shock, the counterfactual implies an additional rise in present-value consumption of about 0.22%p after five years. Given that the baseline model produces a five-year cumulative response of 3.19%p, this counterfactual represents an amplification of about 7.08%.

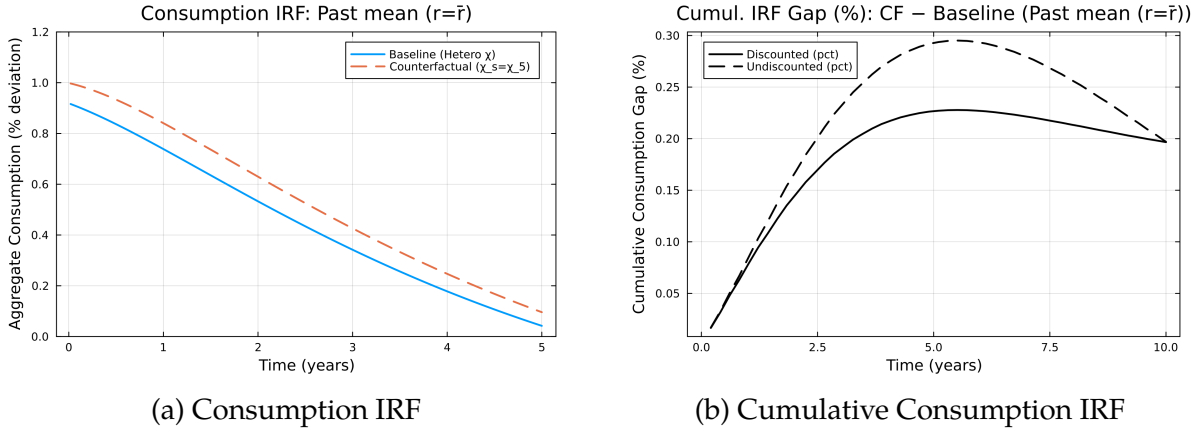


Figure 8: Consumption Response to a 1 pp Rate Cut

It is worth noting that this exercise is consistent with [Berger et al. \(2021\)](#) – the path-dependency of monetary policy. Figure 9 illustrates the case where the past short rate had been 1 percentage point higher than the ergodic mean. Under this scenario, the counterfactual high- χ_{max} economy shows a stronger and more persistent consumption response than the previous past-mean ($r = \bar{r}$) case, with the cumulative gap reaching approximately 0.25%p after five years. Given that the baseline IRF delivers a five-year cumulative response of 3.74%p, this corresponds to an additional consumption effect of about 6.80% percent. The mechanism is straightforward: when the past interest rate path is higher, more households become *in the money* following an identical rate cut, making the attention channel more powerful in amplifying aggregate consumption responses.

Importantly, this counterfactual does not represent a frictionless economy. Even when all households share the same (relatively high) attention level, inattention remains. Yet, the gain in the transmission of monetary policy is already substantial. If overall attention in the population were to rise further, the effect of expansionary monetary policy on aggregate consumption could be even stronger.

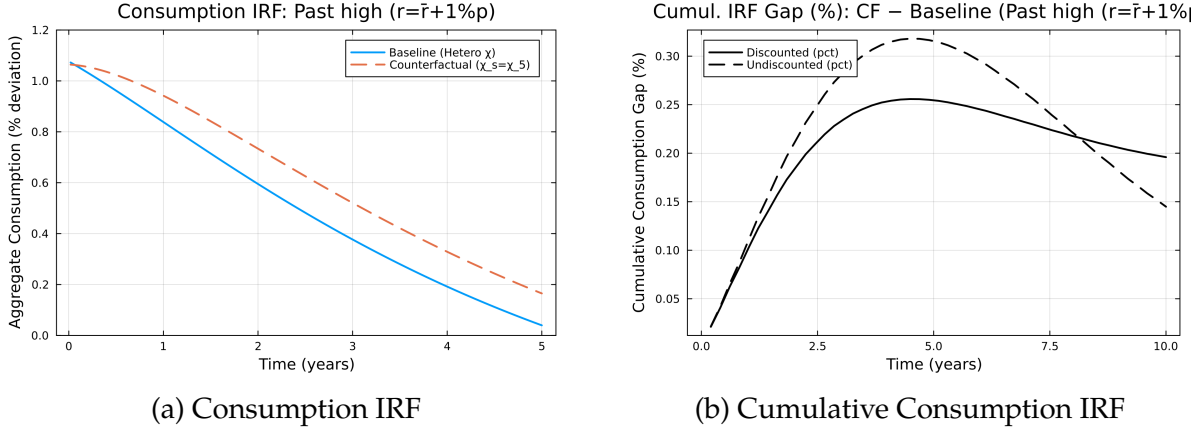


Figure 9: Consumption Response to a 1 pp Rate Cut (Past high rate)

7 Conclusion

This paper presents evidence that heterogeneous refinancing propensities across income groups can significantly impact the transmission effects of monetary policy. Using a broader time series and more robust measures than previous studies, this research shows the low-income households significantly less engage in refinancing inequality than high-income households. Additionally, it demonstrates that the potential increase in disposable income through refinancing is particularly substantial for low-income groups. By analyzing the refinancing response to monetary policy shocks across income quintiles, the study shows that refinancing inequality persists in a dynamic context.

Furthermore, I support the empirical findings with a mortgage refinancing model that incorporates heterogeneous frictions across income groups. The model captures two distinct sources of refinancing inertia: (i) inattention to refinancing opportunities when borrowers are in-the-money, and (ii) execution delays in acting upon profitable refinancing opportunities. The results show that introducing these income-dependent frictions allows the model to replicate the observed pattern of refinancing inequality more accurately than standard homogeneous-friction frameworks. Importantly, the model demonstrates that relaxing attention frictions can substantially amplify the consumption response to an interest rate cut. Given that attention frictions are, in practice, easier to mitigate through policy interventions as shown in the previous studies, these findings highlight a potential channel through which monetary easing could be made more effective.

A Appendix: Tables and Figures

A.1 Figures and Tables

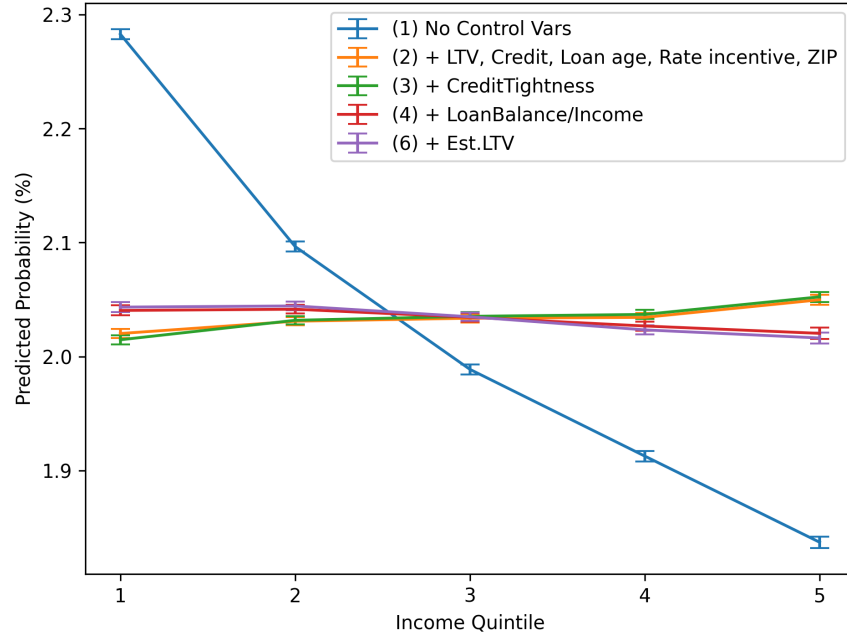


Figure A.1: Predicted Interest Rate Reduction from Refinancing

Notes: This plot shows the interest rate reductions due to refinancing across income quintiles, as estimated by Equation 5. The 95% confidence intervals are represented by error bars and each line corresponds to the columns of the regression result in [Table A.7](#).

A.2 Statistic Moments of the Baseline Model

Table A.6: Mean Values by Income Quintile

Income Quintile	Loan ID (count)	Credit Score	LTV (%)	PTI (%)	Initial Loan Balance (\$)
1	90,331 <24.1>	744.3 (246.7)	69.1 (20.4)	39.5 (8.7)	113,079.9 (48,140.0)
2	78,577 <21.0>	746.9 (201.9)	72.8 (17.5)	36.9 (9.2)	180,094.7 (65,313.4)
3	73,251 <19.5>	750.1 (181.9)	73.4 (16.6)	35.1 (9.6)	236,640.2 (88,343.2)
4	68,386 <18.2>	756.5 (220.8)	73.1 (16.1)	32.7 (9.7)	299,712.5 (113,507.7)
5	64,332 <17.2>	765.6 (235.0)	69.9 (16.4)	25.5 (9.5)	368,110.3 (144,937.8)
Total	374,877 <100.0>	751.8 (219.6)	71.5 (17.8)	34.5 (10.4)	228,178.7 (130,095.4)
Income Quintile	Maturity (Month)	Income (\$)	Monthly Payment (\$)	House Value (\$)	
1	328.8 (66.1)	18,251.7 (5,230.7)	595.0 (205.7)	180,890.9 (106,661.4)	
2	325.2 (68.7)	30,718.6 (4,457.3)	943.4 (263.5)	264,049.8 (132,821.6)	
3	323.6 (70.0)	42,484.3 (5,611.8)	1,239.8 (366.4)	341,449.0 (172,616.7)	
4	319.7 (72.8)	58,208.9 (8,015.1)	1,578.9 (488.9)	431,965.9 (214,241.4)	
5	301.9 (82.9)	103,865.1 (58,277.9)	2,026.4 (715.0)	562,296.5 (305,448.9)	
Total	320.8 (72.3)	47,369.5 (38,036.0)	1,214.4 (654.4)	339,725.3 (232,158.2)	

Notes: The table reports means and standard deviations (in parentheses) by income quintile. The Loan ID column represents the unique number of loans, and the values within the angle brackets indicate the proportion of loans within that quintile relative to the total number of loans. Unconditional income quintiles are divided into 20% segments each, but the matched panel is based on matched refinanced loans. Therefore, the proportion of the matched panel can vary according to the number of refinanced loans in each income quintile.

Table A.7: Interest Rate Reductions by Refinancing

Dependent Equation	Interest Rate Reduction by Refi (%p)				
	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 2	-0.186 ^a (0.00317)	0.0108 ^a (0.00273)	0.0166 ^a (0.00293)	0.000147 (0.00298)	0.000926 (0.00296)
IncomeQuintile 3	-0.294 ^a (0.00320)	0.0134 ^a (0.00283)	0.0216 ^a (0.00307)	-0.00529 ^c (0.00319)	-0.00741 ^b (0.00317)
IncomeQuintile 4	-0.370 ^a (0.00328)	0.0141 ^a (0.00297)	0.0249 ^a (0.00324)	-0.0112 ^a (0.00345)	-0.0176 ^a (0.00343)
IncomeQuintile 5	-0.446 ^a (0.00342)	0.0296 ^a (0.00319)	0.0440 ^a (0.00350)	-0.0133 ^a (0.00402)	-0.0198 ^a (0.00400)
Constant	2.283 ^a (0.00229)	0.963 ^a (0.00701)	0.949 ^a (0.00709)	1.059 ^a (0.00846)	1.045 ^a (0.00843)
<i>Control variables</i>					
LTV, credit score, loan age, ZIP code		x	x	x	x
Credit tightness, HARP			x	x	x
Loan balance to Income				x	x
Est. equity					x
Observations	482,078	481,683	481,683	481,683	481,683
R-squared	0.043	0.351	0.352	0.354	0.361
Equation	Predicted Interest Rate Reduction by Refi (%p)				
	(1)	(2)	(3)	(4)	(5)
IncomeQuintile 1	2.2828 (0.0023)	2.0202 (0.0021)	2.0147 (0.0021)	2.0405 (0.0022)	2.0434 (0.0022)
IncomeQuintile 2	2.0965 (0.0022)	2.0310 (0.0019)	2.0319 (0.0019)	2.0415 (0.0019)	2.0444 (0.0019)
IncomeQuintile 3	1.9888 (0.0022)	2.0337 (0.0019)	2.0353 (0.0019)	2.0345 (0.0019)	2.0349 (0.0019)
IncomeQuintile 4	1.9128 (0.0024)	2.0344 (0.0020)	2.0370 (0.0020)	2.0268 (0.0020)	2.0234 (0.0020)
IncomeQuintile 5	1.8372 (0.0025)	2.0498 (0.0022)	2.0524 (0.0022)	2.0204 (0.0025)	2.0163 (0.0025)

Notes: This table presents estimated interest rate savings through refinancing using [Equation 5](#). The upper section of the table displays the regression results, while the lower section shows predicted values of the dependent variable based on the regression results. Superscripts a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis.

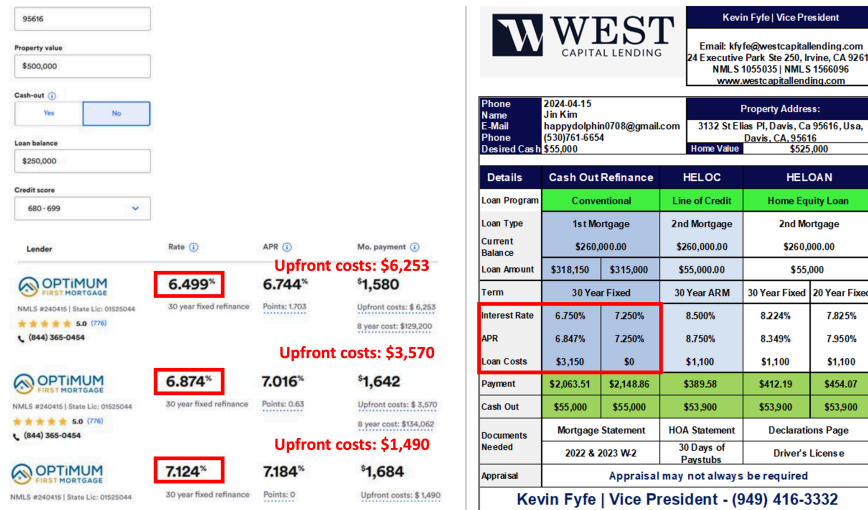


Figure A.2: Predicted Interest Rate Reduction from Refinancing

Notes: This figure shows the offers received from mortgage brokers in a hypothetical scenario for refinancing a home in Davis, California, as of May 2024. The left figure assumes pure rate refinancing (non-cash out), while the right figure assumes a cash-out scenario. Both figures illustrate the tradeoff between upfront costs and interest rates.

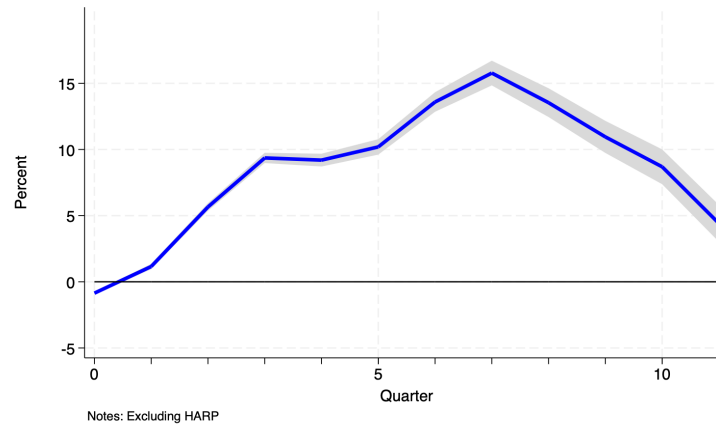


Figure A.3: Cumulative Average Refinancing Response to a 1% Interest Rate Cut

Notes: This plot shows the cumulative refinancing response of all income groups to 1% interest rate falls using Equation 7. The dependent variable is the cumulative proportion of loans refinanced in each region as described in Equation 6. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to 2022 was used. The shading represents the 95% confidence interval.

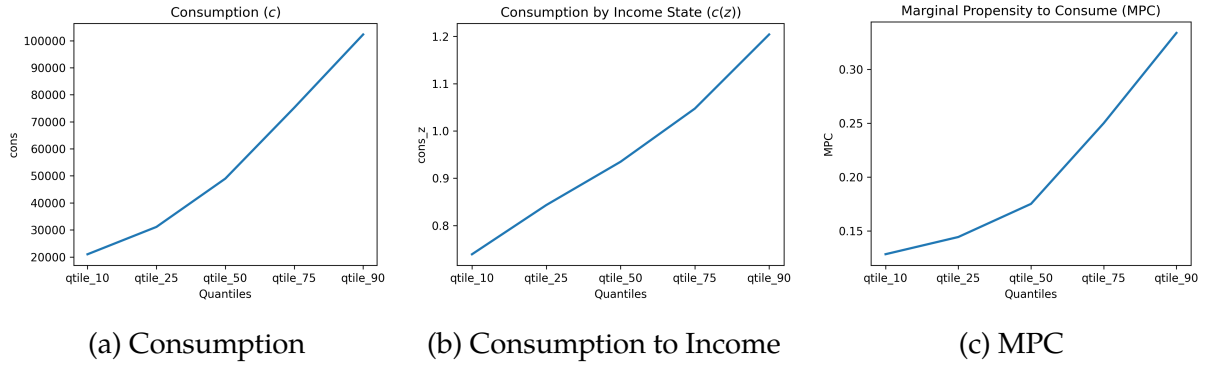


Figure A.4: Comparison across three cases

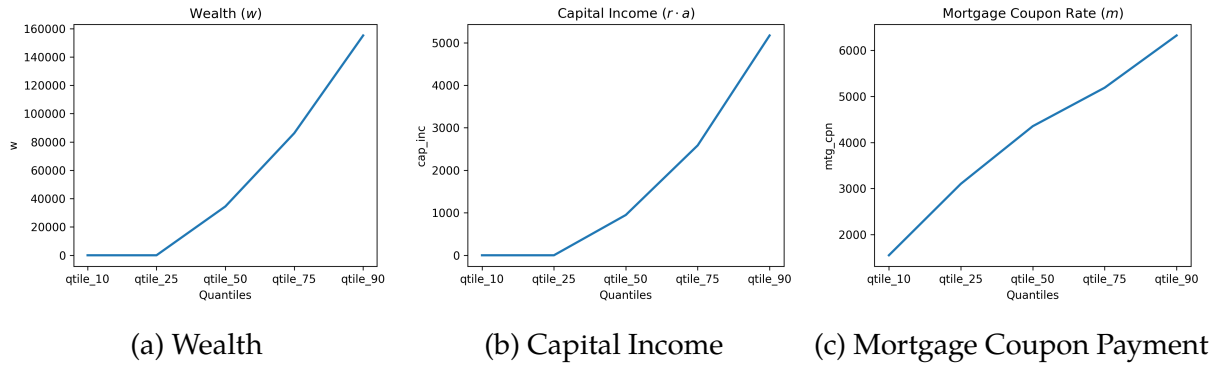


Figure A.5: Cross-Sectional Moments of the Baseline Model

A.3 Data Validity Checks

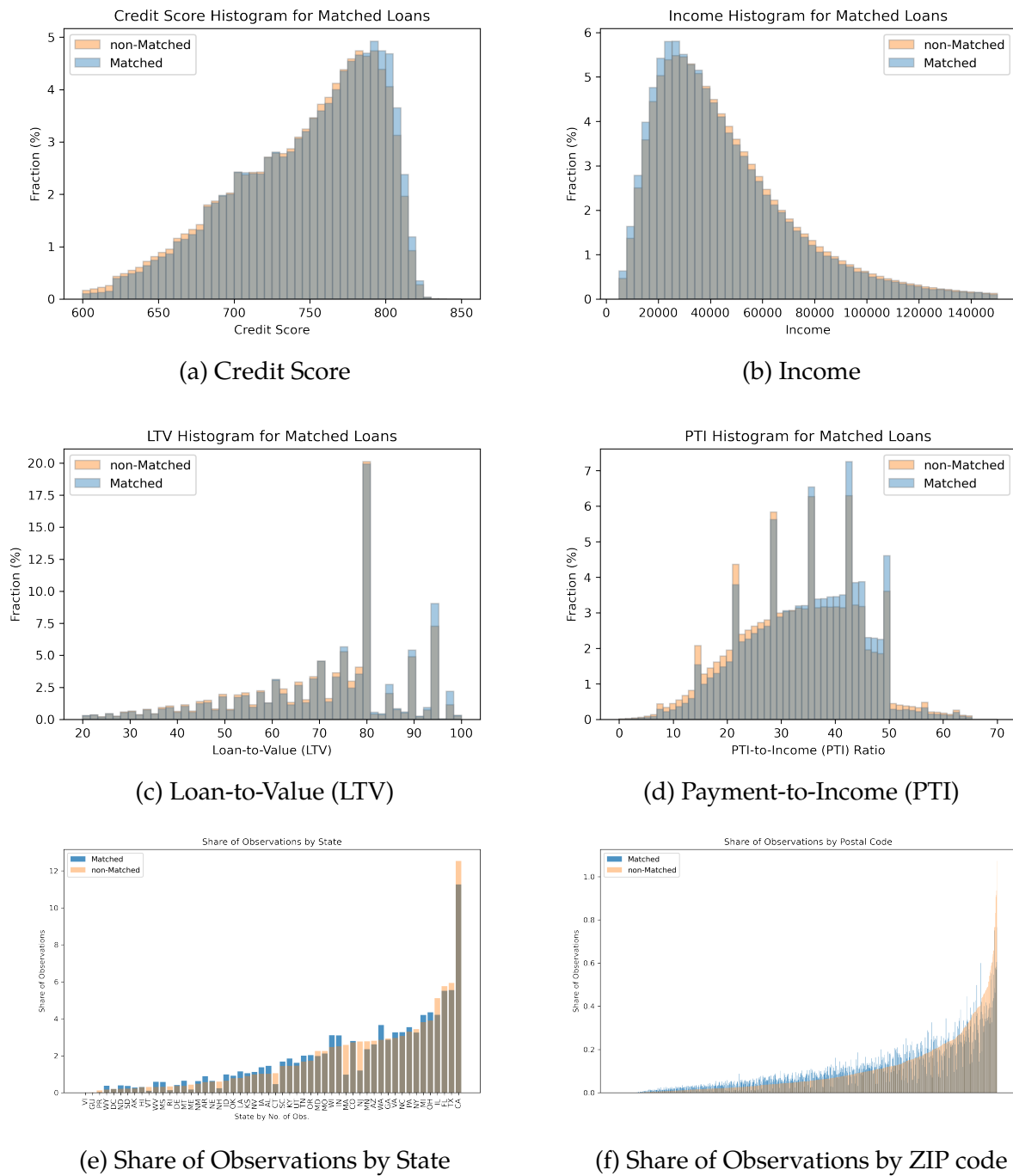


Figure A.6: Matched and Non-Matched Data Comparison by Distribution of Key Variables

Table A.8: Data Validity Checks on the Matched Panel

	Loan ID (count)	Credit Score	Loan Balance	Loan Term	Income	LTV	PTI
Matched Panel (mean)	374,877	751.8	228,179	320.8	47,369	71.5	34.5
All (mean)	41,150,001	758.4	213,022	309.4	49,514	70.1	33.3
Difference	<i>0.9% of total</i>	-0.9%	7.1%	3.7%	-4.3%	2.0%	3.5%
Matched Panel (median)	374,877	757.0	202,000	360.0	38,829	76.0	35.0
All (median)	41,150,001	754.0	184,000	360.0	40,474	75.0	34.0
Difference	<i>0.9% of total</i>	0.4%	9.8%	0.0%	-4.1%	0.0%	2.9%

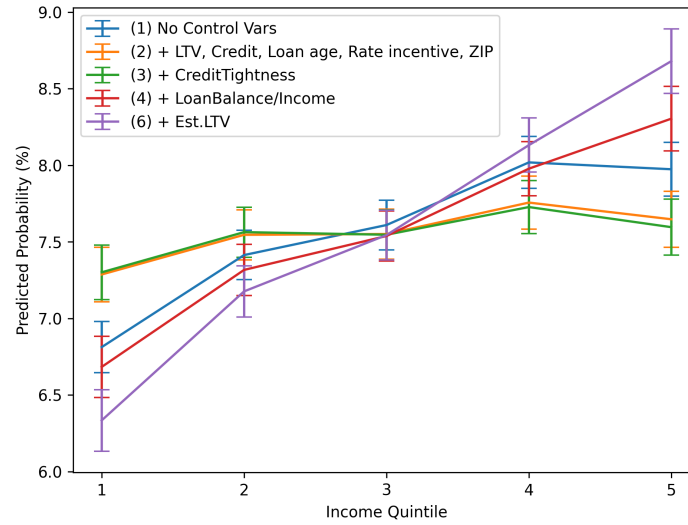
B Appendix: Robustness Checks

B.1 Changes in Income Quintile

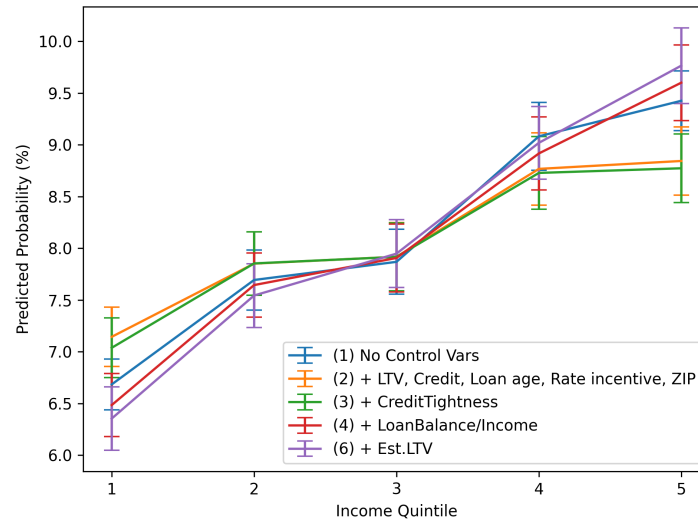
Table B.9: Income Quintile Transition Matrix

Data	Income Quintile	To 1	To 2	To 3	To 4	To 5	Total (%)
PSID ¹	From 1	68.9	14.8	6.8	4.0	5.6	100.0
	2	26.2	35.7	22.6	9.6	6.0	100.0
	3	12.6	17.8	33.2	24.9	11.5	100.0
	4	4.7	7.6	17.2	38.6	31.9	100.0
	5	2.4	1.6	3.2	10.0	82.7	100.0
Loan Panel ²	From 1	43.6	24.6	15.0	10.0	6.8	100.0
	2	20.9	30.4	22.7	15.4	10.7	100.0
	3	9.3	22.4	28.9	23.1	16.3	100.0
	4	4.9	11.4	22.8	33.4	27.5	100.0
	5	2.9	6.1	13.0	25.6	52.4	100.0

Notes: 1) Panel Study of Income Dynamics. Changes of mortgagor's income over two years, 2005-2021 data, 2) Changes from origination to refinancing.



(a) Predicted Refinancing Probability Using New Loan's Information



(b) Predicted Refinancing Probability for Loans Without Income Quintile Changes

Figure B.7: Refinancing Propensity by Income Quintile and Leverage Ratio

Notes: Panel (a) shows the predicted monthly refinancing probability conditional on being in-the-money, using new loan information at the time of refinancing. Panel (b) shows the same probability using samples restricted to loan pairs where the income quintile did not change from origination to refinancing. These regressions are based on Equation 2. Additionally, they are based on matched loans, excluding those that were never refinanced, which leads to a higher monthly refinancing probability than the benchmark regression. The 95% confidence intervals are represented by error bars.

Table B.10: Regression by Loan Age Restrictions

Dependent Loan Age $\leq$	Refinancing Indicator $\times 100$			
	1 yr	2 yrs	3 yrs	5 yrs
IncomeQuintile 2	0.561 ^a (0.245)	1.220 ^a (0.146)	1.229 ^a (0.0951)	0.962 ^a (0.0732)
IncomeQuintile 3	1.235 ^a (0.250)	1.963 ^a (0.153)	2.041 ^a (0.102)	1.741 ^a (0.0783)
IncomeQuintile 4	2.192 ^a (0.259)	2.875 ^a (0.161)	2.752 ^a (0.111)	2.403 ^a (0.0854)
IncomeQuintile 5	2.912 ^a (0.292)	3.642 ^a (0.180)	3.654 ^a (0.127)	3.283 ^a (0.0981)
Constant	0.276 (0.589)	-0.373 (0.354)	-0.244 (0.263)	-0.0230 (0.200)
Observations	97,515	374,996	670,264	945,099
R-squared	0.022	0.015	0.014	0.012

Notes: a, b, c denote significance at 1%, 5%, 10% levels respectively. S.E. in parenthesis. The results base on the benchmark regression (5); control variables are LTV, credit score, rate incentive, credit tightness, HARP, loan balance to income, estimated LTV.

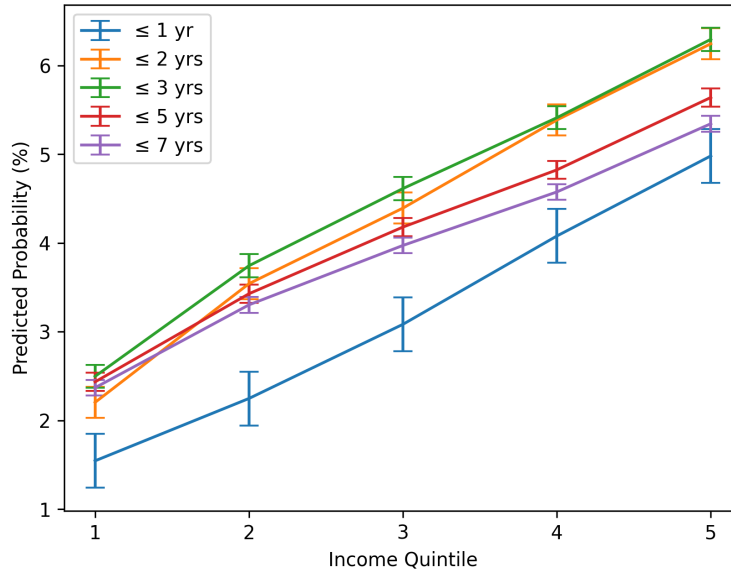


Figure B.8: Predicted Refinancing Probability by Loan Age

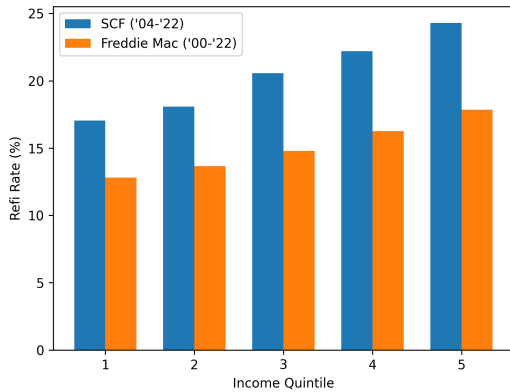
Notes: This plot shows the predicted monthly refinancing probability by loan age, using Equation 2. The results are based on the benchmark regression (5); control variables are LTV, credit score, rate incentive, credit tightness, HARP, loan balance to income, and estimated LTV. The 95% confidence intervals are represented by error bars.

B.2 Survey of Consumer Finances (SCF)

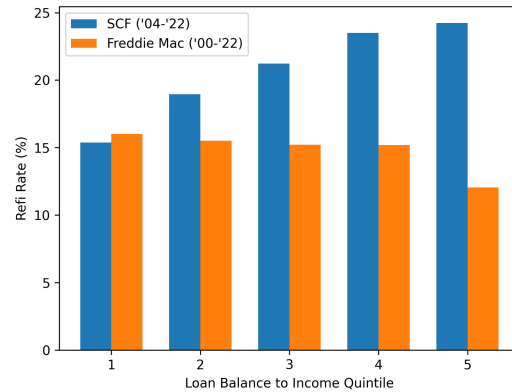
Finally, I utilized the Survey of Consumer Finances (SCF) to examine whether the pattern of refinancing inequality could be observed. While the SCF does not provide panel data and is conducted as a triennial survey, it allows us to observe the year of the most recent refinancing for currently held mortgages. This enables us to derive annual refinancing rates.

Figure B.9 Panel (a) shows the refinancing rates by income quintile, calculated using pooled data from seven surveys conducted between 2004 and 2022. When compared to the Freddie Mac data, the SCF data similarly reveals that lower-income households exhibit a lower propensity to refinance. However, the SCF data overall shows higher refinancing rates, likely due to differences in data coverage, as will be discussed later.

Conversely, Panel (b) displays refinancing rates divided into quintiles based on the leverage (loan balance to income) ratio as per [Chen et al. \(2020\)](#). Interestingly, this presents a contrasting pattern: the SCF data indicates higher refinancing activity as the leverage ratio increases, whereas the Freddie Mac data shows a decrease in refinancing propensity.



(a) Refinancing Rate by Income



(b) Refinancing Rate by Leverage Ratio

Figure B.9: Refinancing Propensity by Income Quintile and Leverage Ratio

The divergence in these results can be attributed to two main factors. First, the nature of the datasets differs significantly. The SCF is a survey-based triennial dataset that uses current household income, whereas the Freddie Mac data is an administrative panel dataset based on the borrower's income at the time of loan origination. Consequently, differences in the calculation of the loan-to-income ratio, due to vari-

ations between household and individual income and current versus past income, complicate direct comparisons.

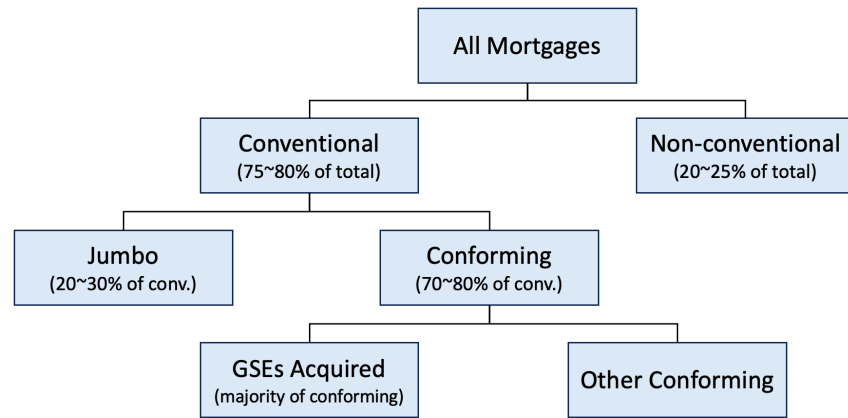


Figure B.10: Mortgage Types in the U.S.

Notes: Market shares of each mortgage type vary over time, and the numbers provided are approximate, referencing the National Mortgage Database (NMDb), web searches.

More importantly, the scope of households covered by each dataset appears to contribute to these contrasting results. As shown in [Figure B.10](#), the types of mortgages in the U.S. are broadly categorized into conventional and non-conventional mortgages. Conventional mortgages are further divided into conforming and jumbo (non-conforming) loans. Conforming loans meet the guidelines set by the Federal Housing Finance Agency (FHFA), including loan limits and credit requirements, whereas jumbo loans exceed these limits. Conforming loans are mostly acquired in the secondary market by government-sponsored enterprises (GSEs) like Fannie Mae and Freddie Mac. The Freddie Mac data that I used primarily represents about 70% of conventional loans, accounting for approximately 60% of all loans.

However, the loans not covered by the Freddie Mac data, particularly the jumbo loans, which constitute about 30% of conventional loans, seem to be influencing the results. Jumbo loans, exceeding conforming loan limits, are predominantly held by high-income, wealthy individuals. Since the SCF samples all households, including jumbo loan holders, it exhibits significantly higher income levels, even considering household versus borrower income as shown in [Table B.11](#).

Thus, assuming comparability despite the inherent differences in data characteristics, the sample including all households shows increased refinancing with higher leverage. In contrast, narrowing the analysis to middle and lower-income households reveals an opposite and intriguing pattern. Highly leveraged households might have

Table B.11: Freddie Mac (Borrower) and SCF (Household) Income by Leverage Ratio

Loan-to-income Quintile	Median Income (\$)		Mean Income (\$)	
	Freddie Mac	SCF	Freddie Mac	SCF
1	53,143	356,204	68,713	1,790,221
2	41,095	161,175	47,831	366,883
3	35,438	119,000	40,494	189,615
4	31,529	95,000	35,493	142,475
5	28,984	58,396	31,971	83,983

Notes: It is important to note that the SCF uses current household income, while Freddie Mac uses the borrower's individual income at the time of loan origination. The statistics are based on the SCF 2004-2022 triannual public data by pooling it together, and Freddie Mac used a sample based on loans originated from 2000 to 2022.

a high demand for cash-out refinancing due to idiosyncratic shocks, but those with higher absolute incomes might experience fewer frictions related to low income, such as liquidity constraints for closing costs or financial illiteracy, which affect refinancing decisions.

Regarding policy relevance, which dataset is more relevant? If the limitations of survey data is disregard, the SCF offers a more comprehensive coverage. However, if monetary policy focuses on relatively liquidity-constrained, high-MPC middle and lower-income households, the subset I used would be more significant for policymakers. Specifically, frictions unique to lower-income households can disrupt the transmission channels of monetary policy, making the refinancing behavior of the wealthy less pertinent. Comparing refinancing propensity using consistent income standards across different data scopes could be an interesting area for further research.

B.3 Different Specifications of Refinancing Response

Apart from Equation 6, there may be other specifications to measure regional refinancing responses. Therefore, in this section, I verify robustness by using alternative methods.

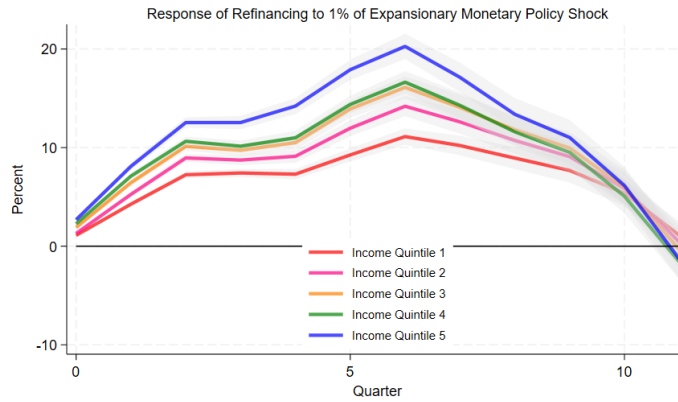
$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q} + \dots + x_{t+h}^{r,q}}{y_{t-1}^{r,q}} \quad (18)$$

$$\rho_{t+h}^{r,q} = \frac{x_t^{r,q}}{y_t^{r,q}} + \dots + \frac{x_{t+h}^{r,q}}{y_{t+h}^{r,q}} \quad (19)$$

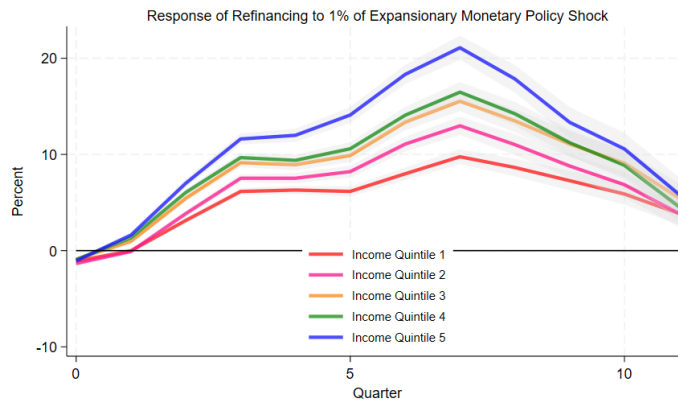
Again, $x_t^{r,q}$ is the number of refinancing loans by income quintile q in region r in quarter t , $y_t^{r,q}$ is the number of all outstanding loans, and h is the analysis time horizon.

Equation 18 fixes the denominator to the period before the monetary shock arrives. This approach reduces the distortion caused by underestimating the refinancing share due to an increase in the total number of outstanding loans prompted by home purchases in that region. On the other hand, if the horizon extends and there are loans that refinance multiple times, it may overestimate the response magnitude. Equation 19 is the cumulative sum of the refinancing ratios for each quarter. If the total number of outstanding mortgages does not change significantly, it is expected to show a ratio similar to Equation 18.

Both specifications consistently demonstrate refinancing inequality. Although the timing of the peak differs by only one quarter, both indicate that high-income households refinance significantly more than low-income households.



(a) Equation 18



(b) Equation 19

Figure B.11: Refinancing Response to 1% Interest Rate Cut by Income Quintile

Notes: Panel (a) shows the cumulative refinancing response by income quintile to 1% interest rate falls using Equation 19. The dependent variable is the cumulative proportion of loans refinanced in each region and income group as described in Equation 5. Regions are 908 three-digit ZIP code, and quarterly data from 2004 to 2022 was used

C Appendix: Identification of In-the-money Loans

As described in the text, an in-the-money loan refers to a situation where the difference between the old and new interest rates after refinancing exceeds a certain threshold. This threshold represents the fixed costs associated with refinancing and option value.

$$\text{In-the-money if } \underbrace{(\text{Interest Rate}_i^{\text{old}} - \text{Potential Refi Rate}_{i,t}^{\text{new}}) - \text{Threshold}_{i,t}}_{\equiv \text{Rate Incentive}} > 0$$

(i and t denote loan and time respectively)

Agarwal et al. (2013) has proposed the following tractable closed-form solution for the threshold value.

$$\begin{aligned} \text{Threshold} &= \frac{1}{\psi} [\phi + W(-\exp(-\phi))] \\ \psi &= \sqrt{\frac{2(\rho + \lambda)}{\sigma}} \\ \phi &= 1 + \psi(\rho + \lambda) \frac{\kappa(M)/M}{(1 - \tau)} \end{aligned}$$

Here, $W(\cdot)$ is the Lambert W-function, ρ is the real discount rate, λ is the expected real rate of exogenous mortgage repayment, σ is the standard deviation of the mortgage rate, $\kappa(M)/M$ is the ratio of the tax-adjusted refinancing cost and the remaining mortgage value, and τ is the marginal tax rate. The values of the parameters suggested by Agarwal et al., 2013 are $\sigma = 0.0109$, $\tau = 0.28$, $\rho = 0.05$.

$\kappa(M)$ is the cost entailed in refinancing, which is defined by:

$$\kappa(M) = F + fM \left[1 - \frac{\tau}{\theta + \rho + \pi} \left\{ \frac{1 - \exp(-(\theta + \rho + \pi)N)}{N} \frac{\rho + \pi}{\theta + \rho + \pi} + \theta \right\} \right]$$

F is the fixed cost of refinance, f is the points divided by 100, τ is the marginal tax rate, θ is the expected arrival rate of a full deductibility event, such as moving and refinancing. Using the parameter suggested by the paper $F = 2000$, $f = 0.01$, $\theta = \mu + 0.1 = 0.2$, the cost can be approximated as follows.

$$\kappa(M) = 2000 + 0.007905M$$

Lastly, Lambda is defined as the expected rate of decline in the real principal of the mortgage for reasons other than rate-reducing refinancing, and is defined as:

$$\lambda = \mu + \frac{i^{old}}{\exp(i^{old}\Gamma) - 1} + \pi,$$

Γ is the remaining life of the existing mortgage in years, i^{old} is the current (old) mortgage interest rate for existing loan, μ is the hazard of relocation, and π is the inflation rate. The parameter values used are $\mu = 0.1$ and $\pi = 0.03$, following the paper.

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